NeuraEdge Universal: Deep Debug Plots and Waveforms

Generated from CIFAR-10, XSim, compiler, and Vivado artifacts

July 16, 2026

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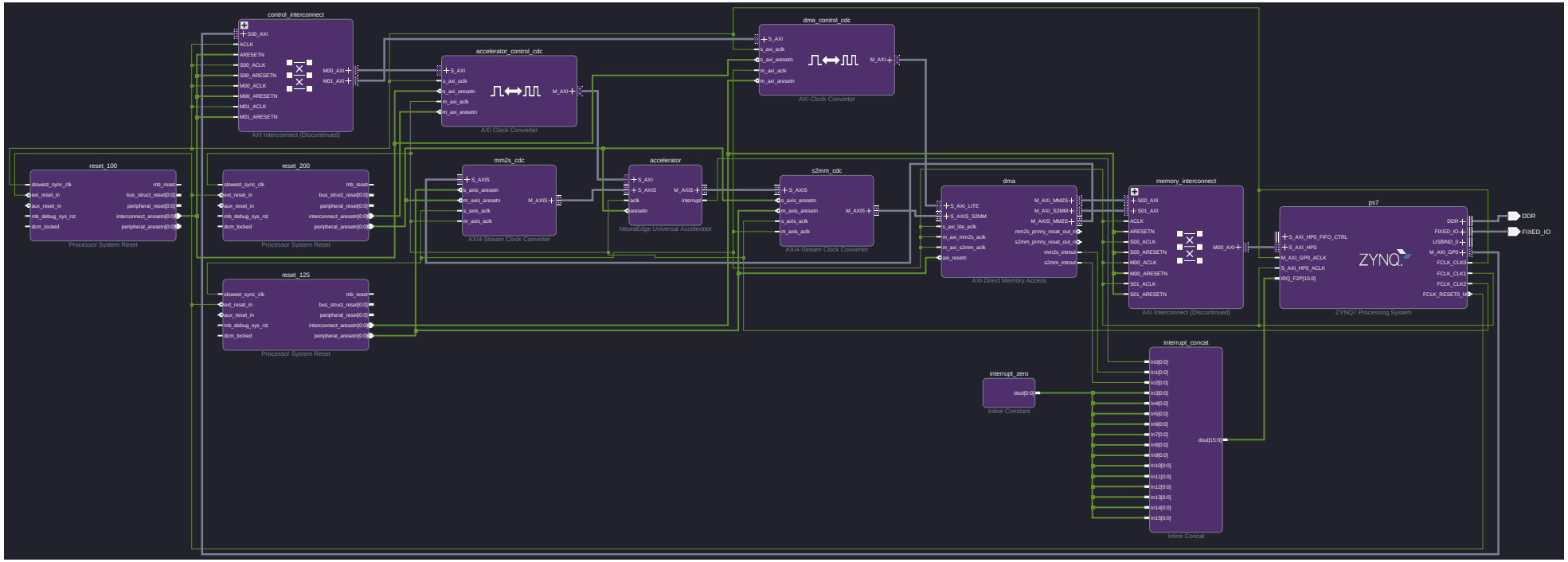
51 Routed timing path distribution **54**

1 Evidence rule

Every page names its source artifact. CIFAR-10 plots use all 10,000 official test samples. Waveforms are generated by Vivado 2026.1 XSim from repository testbenches. Timing and power pages are included only when the corresponding routed report exists. A Vivado estimate is not relabeled as a physical board measurement.

2 Native Vivado PYNQ block design

This page is exported directly from the validated system.bd block design by Vivado 2026.1. It is not a redrawn or inferred schematic.


control_interconnect
S00_AXI
ACLK
ARESETN
S00_ACLK
S00_ARESETN
M00_ACLK
M00_ARESETN
M01_ACLK
M01_ARESETN
M00_AXI
M01_AXI
AXI Interconnect (Discontinued)
accelerator_control_cdc
S_AXI
s_axi_aclk
s_axi_aresetn
m_axi_aclk
m_axi_aresetn
M_AXI
AXI Clock Converter
dma_control_cdc
S_AXI
s_axi_aclk
s_axi_aresetn
m_axi_aclk
m_axi_aresetn
M_AXI
AXI Clock Converter
reset_100
slowest_sync_clk
ext_reset_in
aux_reset_in
mb_debug_sys_rst
dcm_locked
mb_reset
bus_struct_reset[0:0]
peripheral_reset[0:0]
interconnect_aresetn[0:0]
peripheral_aresetn[0:0]
Processor System Reset
reset_200
Processor System Reset
reset_125
Processor System Reset
mm2s_cdc
S_AXIS
s_axis_aresetn
m_axis_aresetn
s_axis_aclk
m_axis_aclk
M_AXIS
AXI4-Stream Clock Converter
accelerator
S_AXI
S_AXIS
aclk
aresetn
M_AXIS
interrupt
NeuraEdge Universal Accelerator
s2mm_cdc
S_AXIS
s_axis_aresetn
m_axis_aresetn
s_axis_aclk
m_axis_aclk
M_AXIS
AXI4-Stream Clock Converter
dma
S_AXI_LITE
S_AXIS_S2MM
s_axi_lite_aclk
m_axi_mm2s_aclk
m_axi_s2mm_aclk
axi_resetn
M_AXI_MM2S
M_AXI_S2MM
M_AXIS_MM2S
mm2s_prmry_reset_out_n
s2mm_prmry_reset_out_n
mm2s_introut
s2mm_introut
AXI Direct Memory Access
memory_interconnect
S00_AXI
S01_AXI
ACLK
ARESETN
S00_ACLK
S00_ARESETN
M00_ACLK
M00_ARESETN
S01_ACLK
S01_ARESETN
M00_AXI
AXI Interconnect (Discontinued)
ps7
S_AXI_HP0_FIFO_CTRL
S_AXI_HP0
M_AXI_GP0_ACLK
S_AXI_HP0_ACLK
IRQ_F2P[15:0]
DDR
FIXED_IO
USBIND_0
M_AXI_GP0
FCLK_CLK0
FCLK_CLK1
FCLK_CLK2
FCLK_RESET0_N
ZYNQ7 Processing System
DDR
FIXED_IO
interrupt_zero
dout[0:0]
Inline Constant
interrupt_concat
in0[0:0]
in1[0:0]
in2[0:0]
in3[0:0]
in4[0:0]
in5[0:0]
in6[0:0]
in7[0:0]
in8[0:0]
in9[0:0]
in10[0:0]
in11[0:0]
in12[0:0]
in13[0:0]
in14[0:0]
in15[0:0]
dout[15:0]
Inline Concat

3 Measured evidence summary

Evidence: evaluation JSON, execution manifest, parameter manifest

Measured Evidence Summary

```
Dataset: CIFAR-10 official test split (10,000 samples)

Checkpoint: 9,555/10,000 = 95.55%

Exported integer arithmetic: 9,542/10,000 = 95.42%

Prediction disagreements: 257

Model MAC equivalents: 3,732,164,352

Tile commands: 104,256

Hardware parameter archive: 9,385,704 bytes, SHA-256 b02c5df519eaecffb228f4c8be21babddc01761f26210bb8dfcca3318ed0d381

Power pages use Vivado report data only; no physical-board value is inferred.

Full PYNQ routed setup/hold slack: +0.005 / +0.050 ns

Clock domains: accelerator 200 MHz, DMA 125 MHz, control 100 MHz

Full-system SAIF estimate: 1.937 W total, 1.785 W dynamic, confidence Medium, nets matched 7%   (9164/123250)
```

4 Complete exported-graph trace validation

Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\regression\real_full_graph_forward_validation.json

Complete Exported-Graph Trace Validation

Result: PASS

Official CIFAR-10 fixture: outputs/cifar10_shards/cifar10_test_000.npz:0

Label / predicted class: 3 / 3

Complete graph commands: 1,345,393

Input / output stream beats: 41,408,322 / 13,105,354

Maximum logit delta from archived 10,000-image evaluation: 9.596e-08

Parameter bytes / SHA-256: 9,385,704 / b02c5df519eaecffb228f4c8be21babddc01761f26210bb8dfcca3318ed0d381

Host trace elapsed time: 197.466 s

5 Final PYNQ artifact integrity

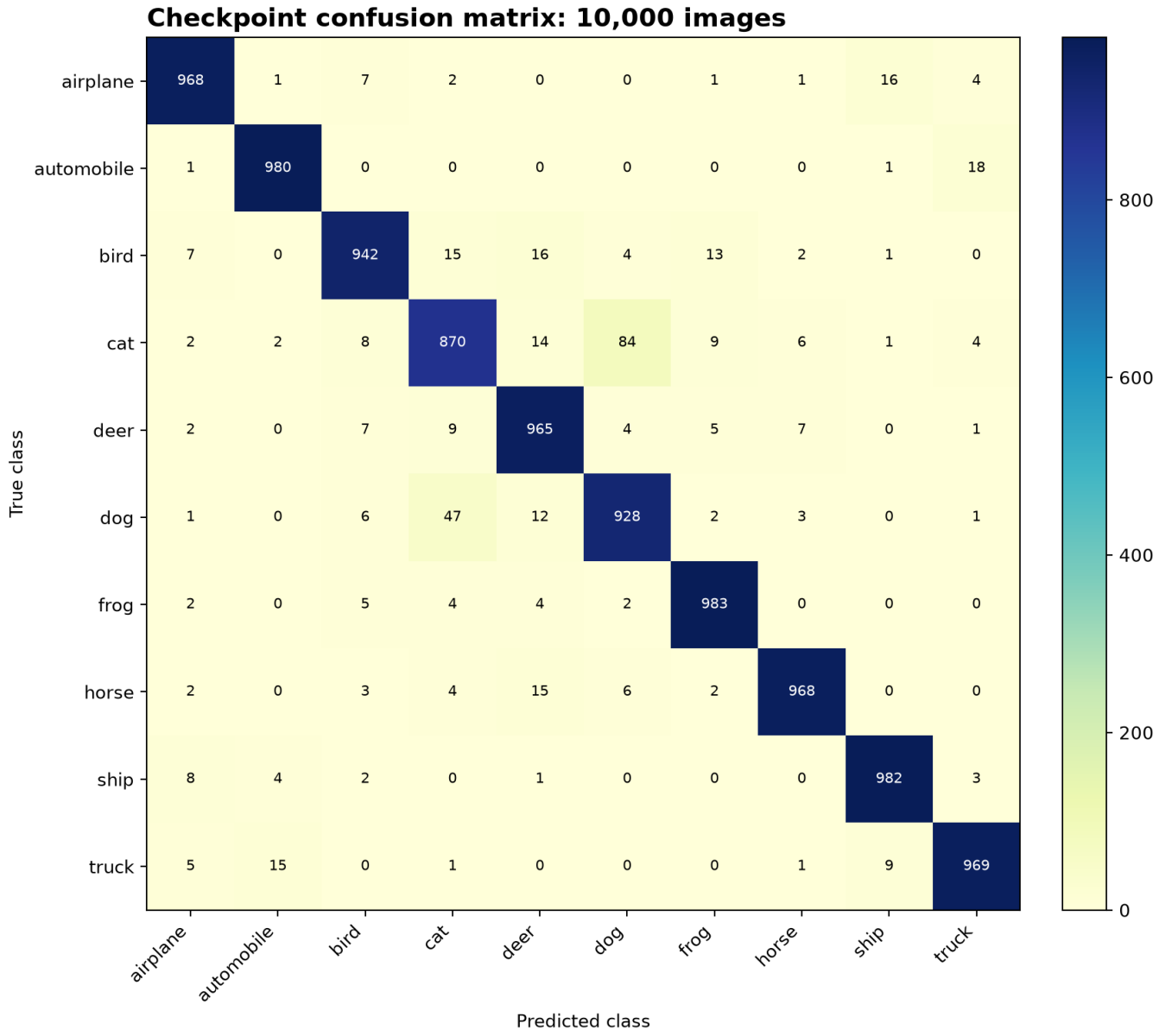
Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\pynq_full

Final PYNQ Deliverable Integrity

```
neuraedge_pynq_full.bit: 4,045,769 bytes
  SHA-256 d9bcac6224efd44da8aa2f8a0e674f5e66a67294caac460d6cfc8086d5a341de
neuraedge_pynq_full.hwh: 384,395 bytes
  SHA-256 79395d99357fa798ecd31b0c9229500dce7e7aada8185c3576b744fd4485712f
neuraedge_pynq_full.xsa: 2,814,632 bytes
  SHA-256 8d06b8a0b6e82105992cbf05fd1f0d3cfac8f5316d682f976f88221b0583bd64
neuraedge_pynq_full_routed.dcp: 32,256,514 bytes
  SHA-256 750e7185560d3187c7576f84b722918d3c8d3da21738d417b8fa59fc9ca857a1
```

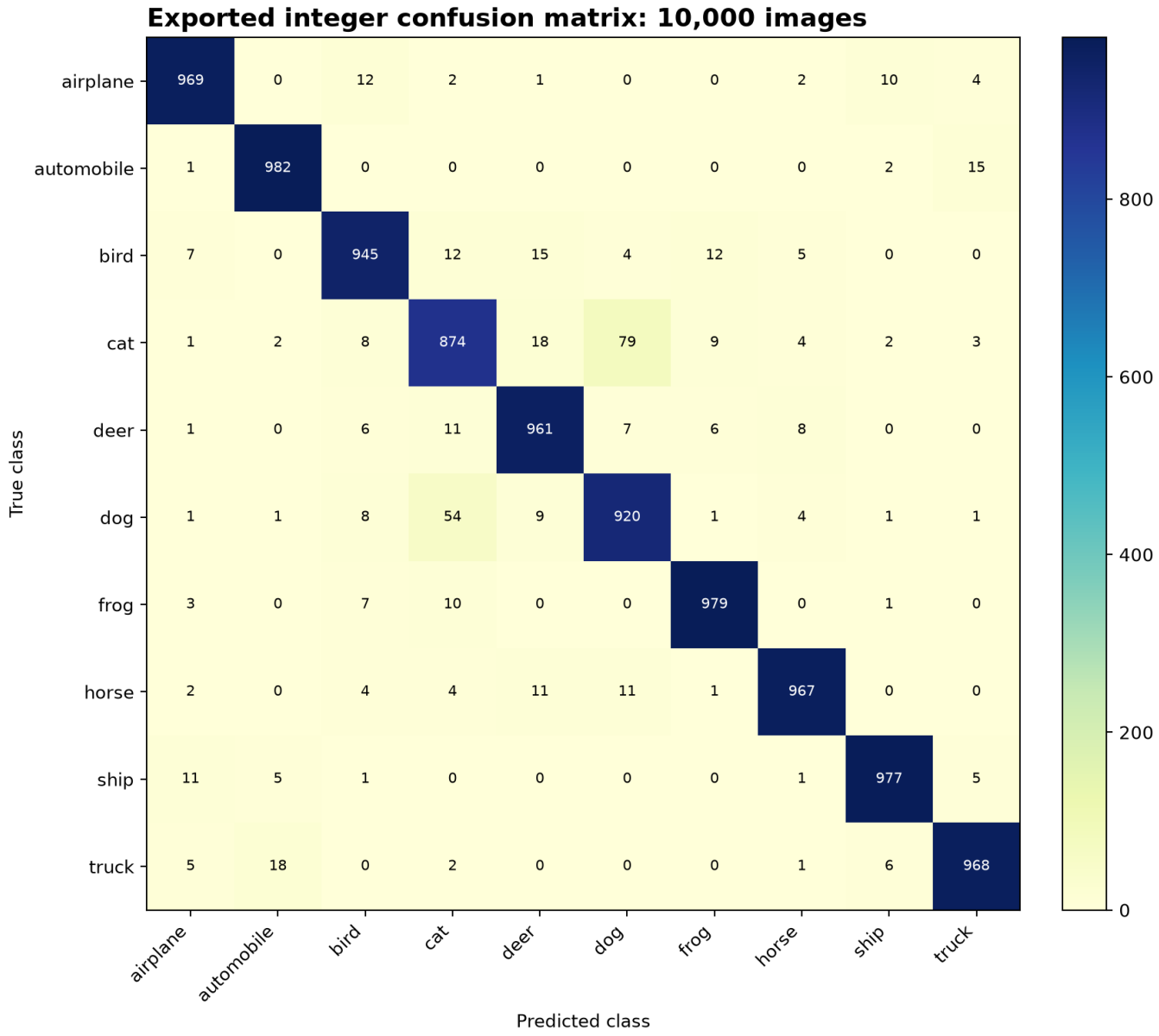
6 Checkpoint confusion matrix: 10,000 images

Evidence: full CIFAR-10 prediction NPZ/JSON



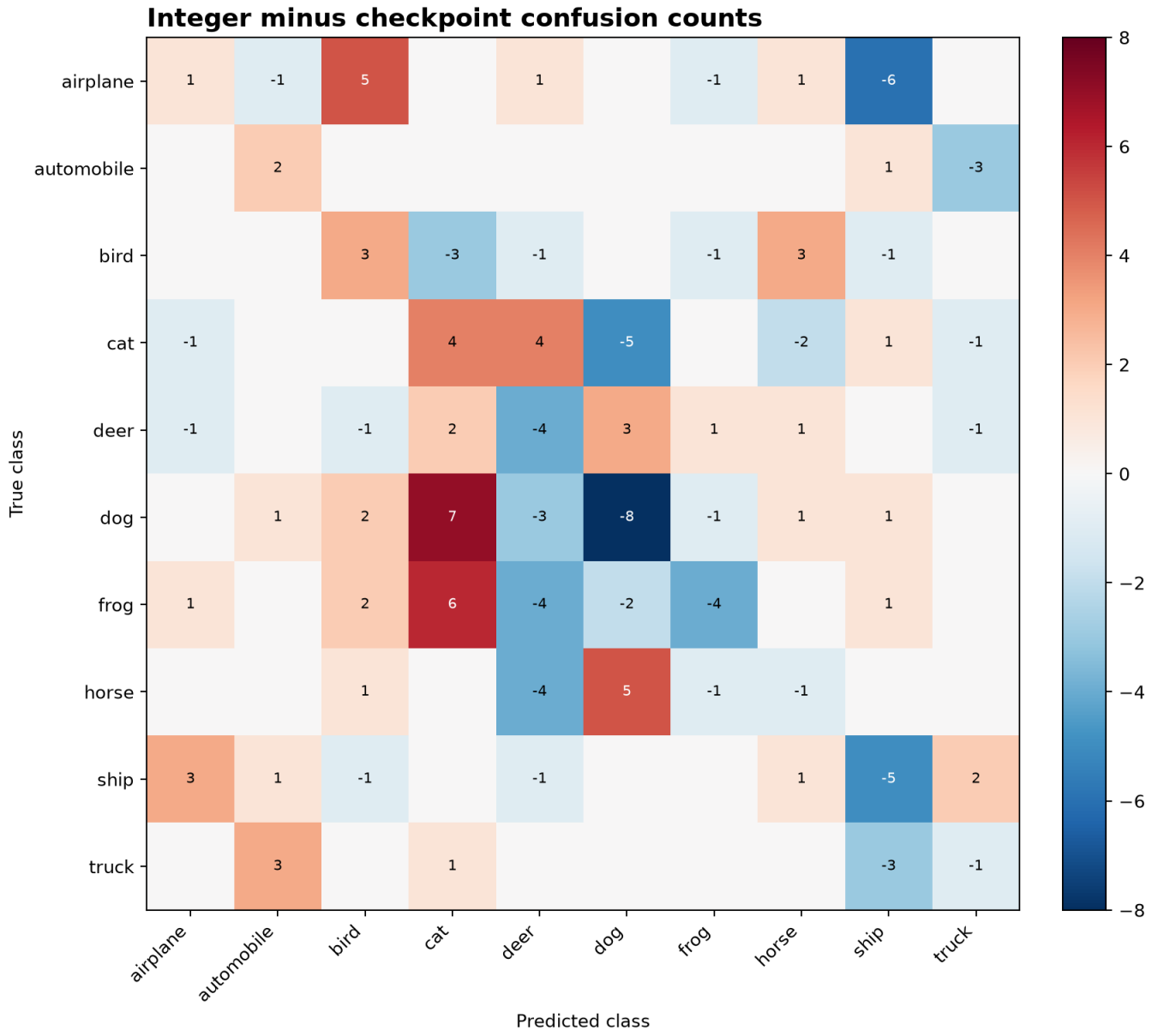
7 Exported integer confusion matrix: 10,000 images

Evidence: full CIFAR-10 prediction NPZ/JSON



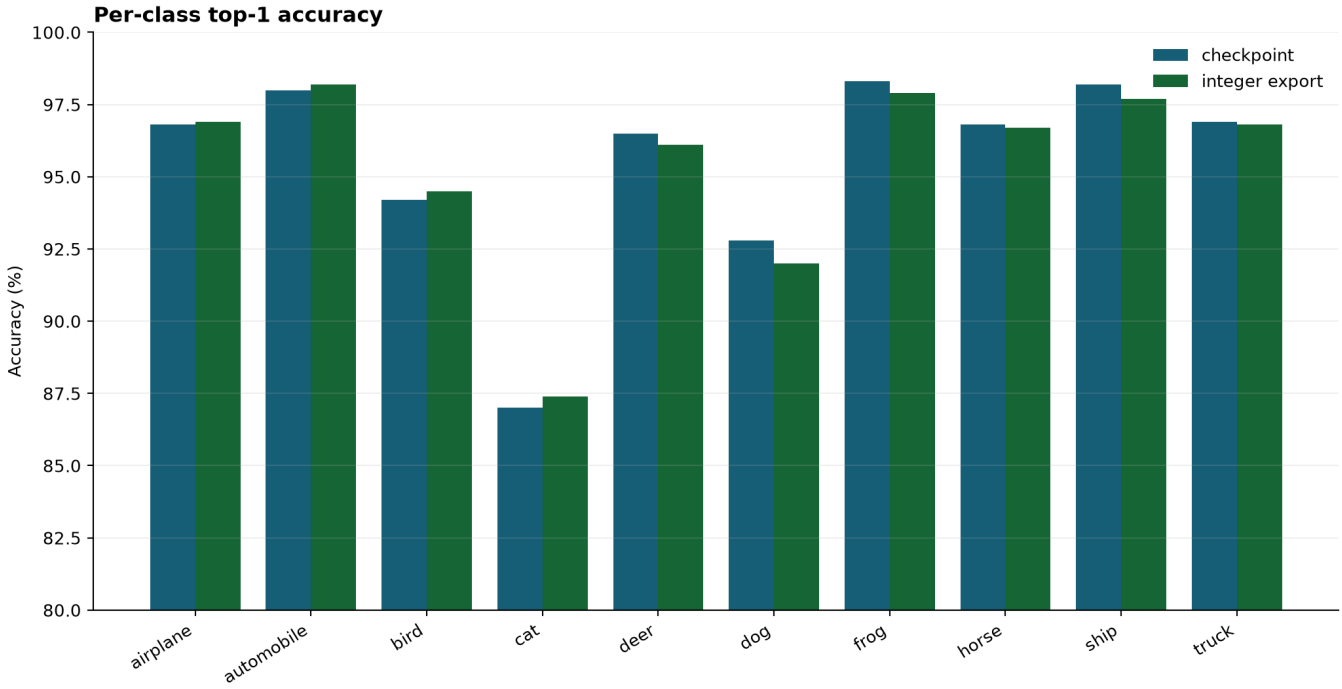
8 Integer minus checkpoint confusion counts

Evidence: full CIFAR-10 prediction NPZ/JSON



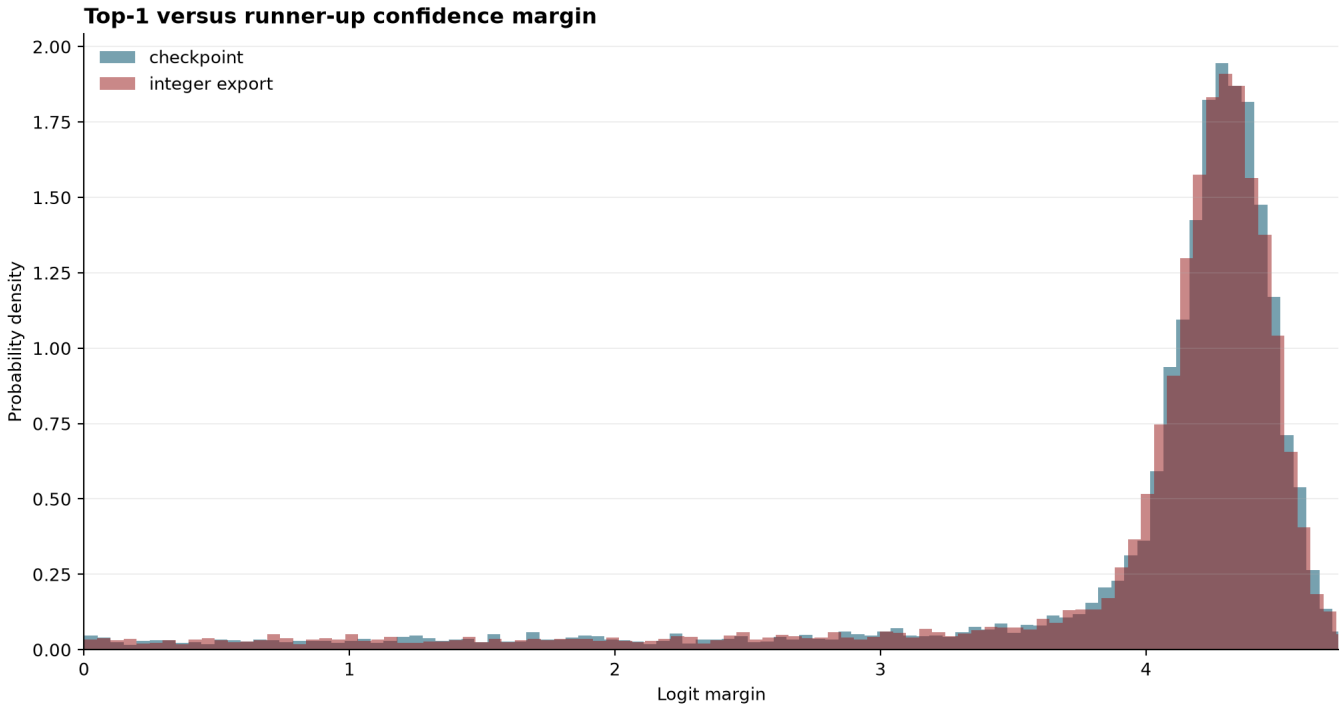
9 Per-class top-1 accuracy

Evidence: full CIFAR-10 confusion matrices



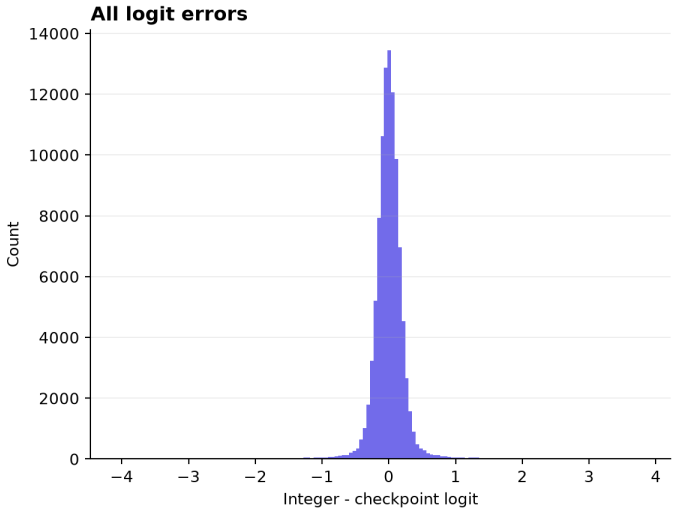
10 Classifier margin distribution

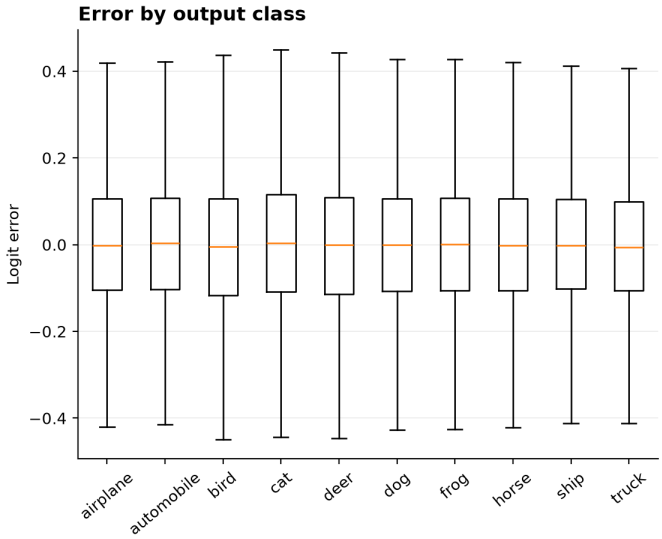
Evidence: all 10,000 checkpoint and integer logits



11 Integer-export logit error

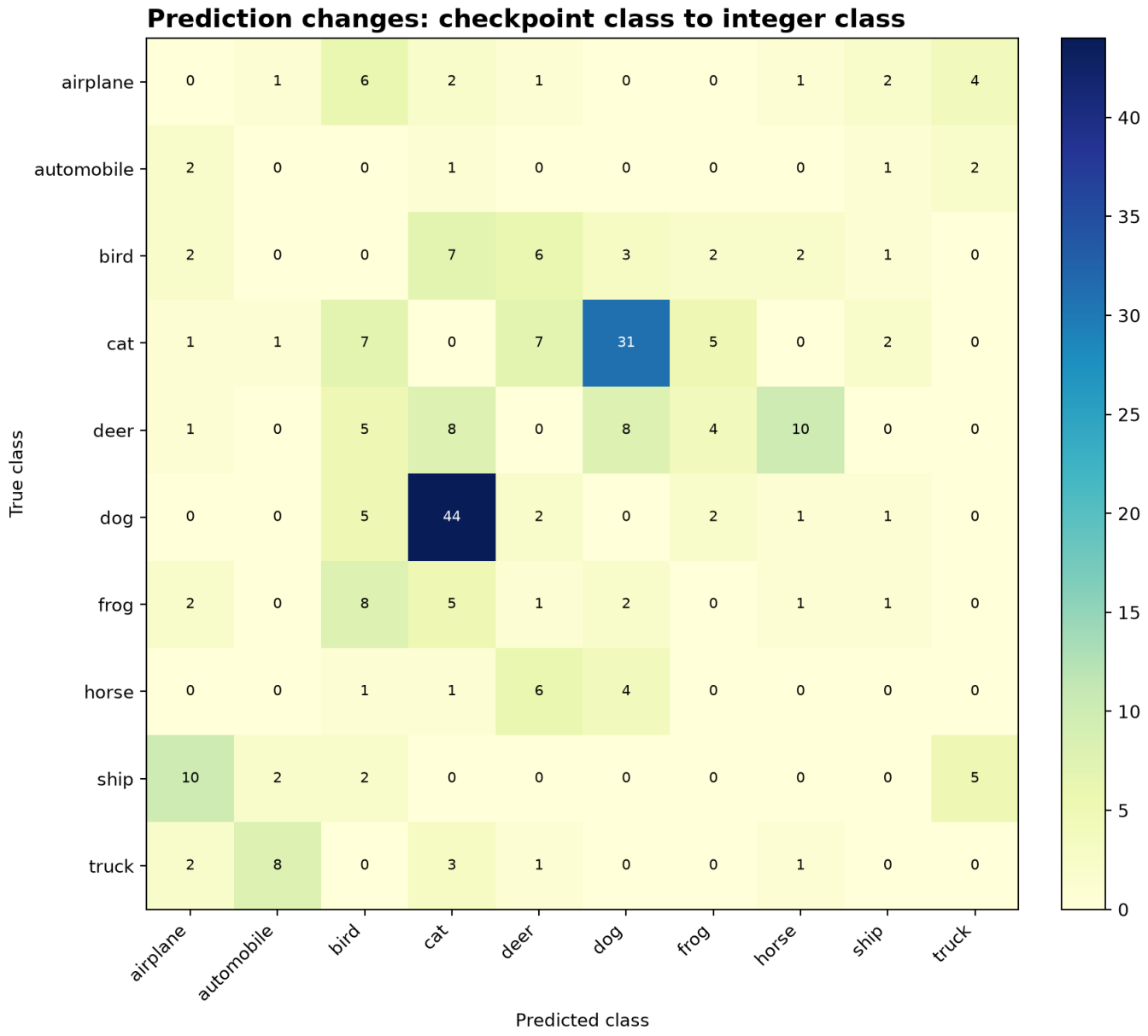
Evidence: paired 10,000 x 10 logit arrays





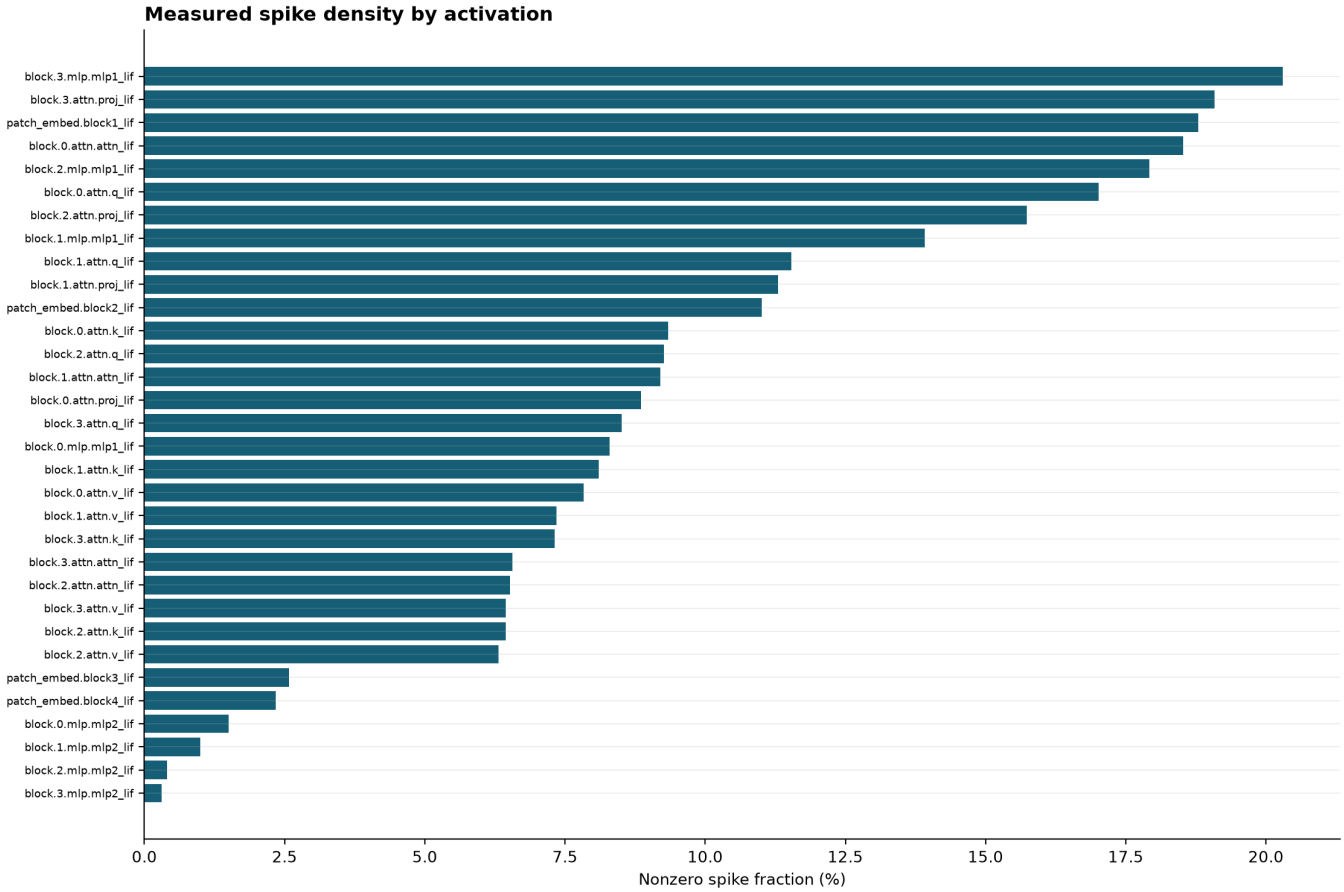
12 Prediction disagreement transitions

Evidence: 257 changed predictions from paired NPZ files



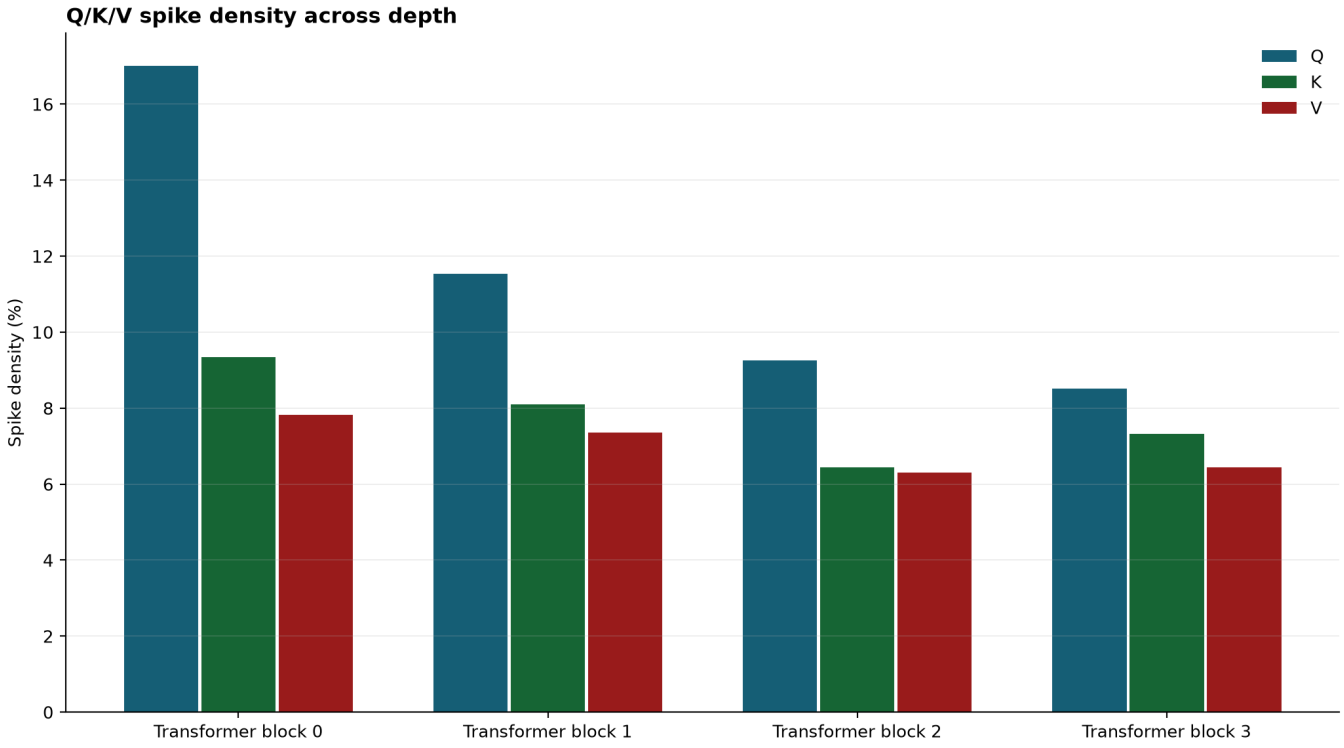
13 Measured spike density

Evidence: streaming activation statistics over all CIFAR-10 test images



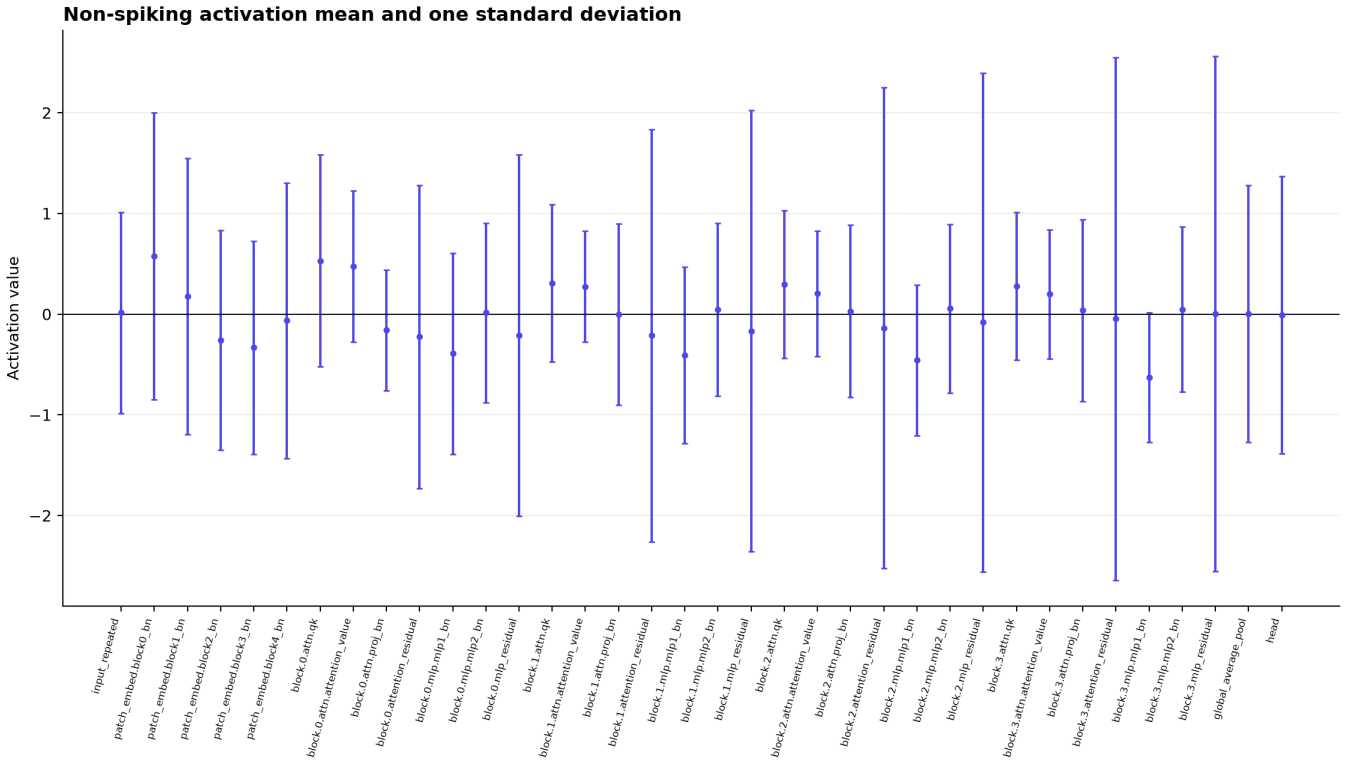
14 Q/K/V spike density

Evidence: full-test streaming activation statistics



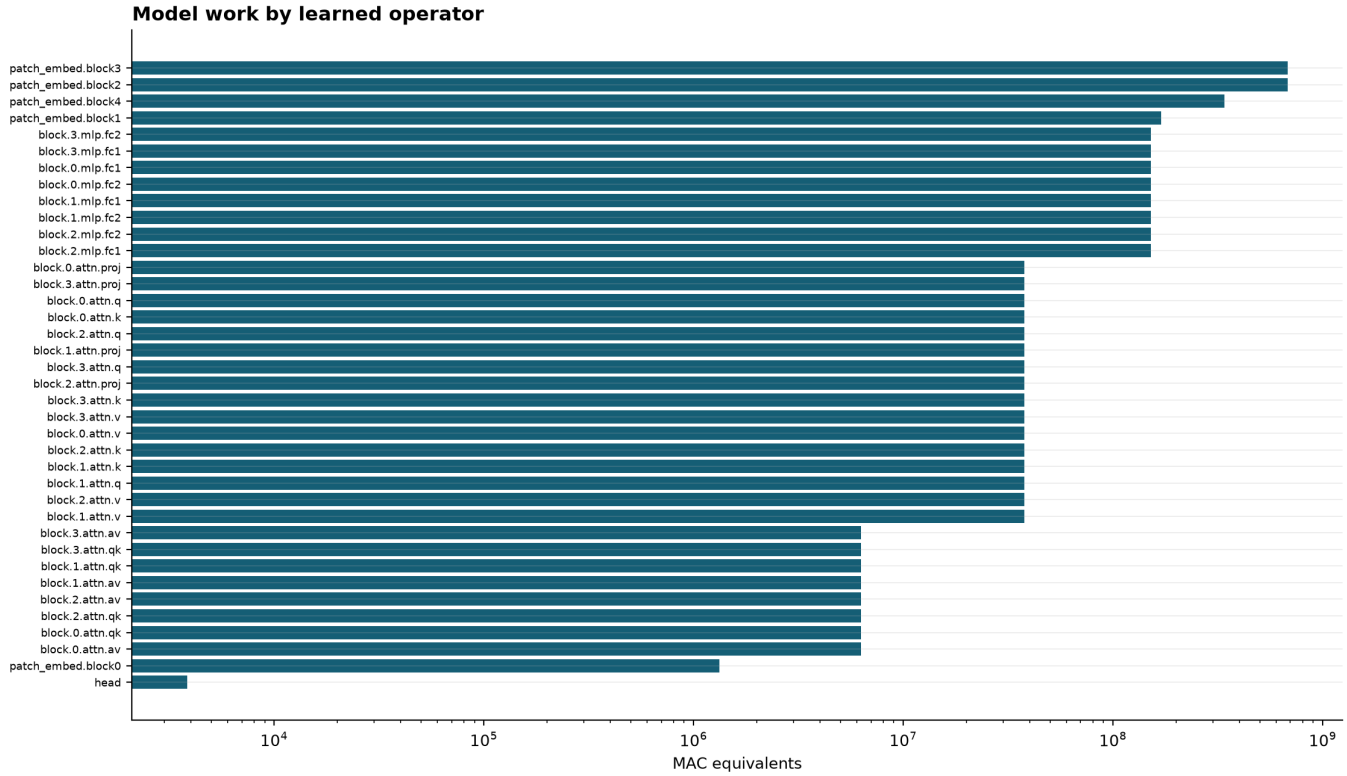
15 Activation mean and spread

Evidence: full-test streaming activation moments



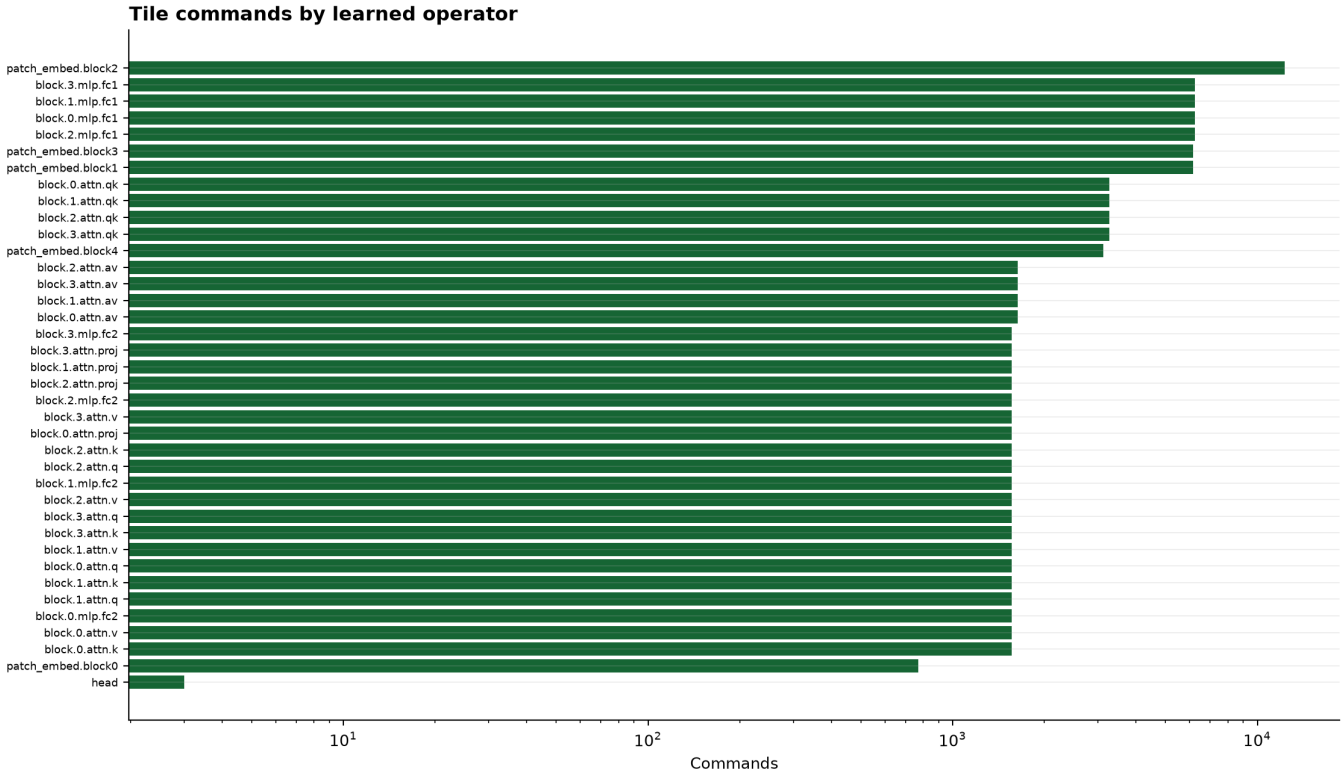
16 Model work by learned operator

Evidence: execution manifest generated from exported tensor shapes



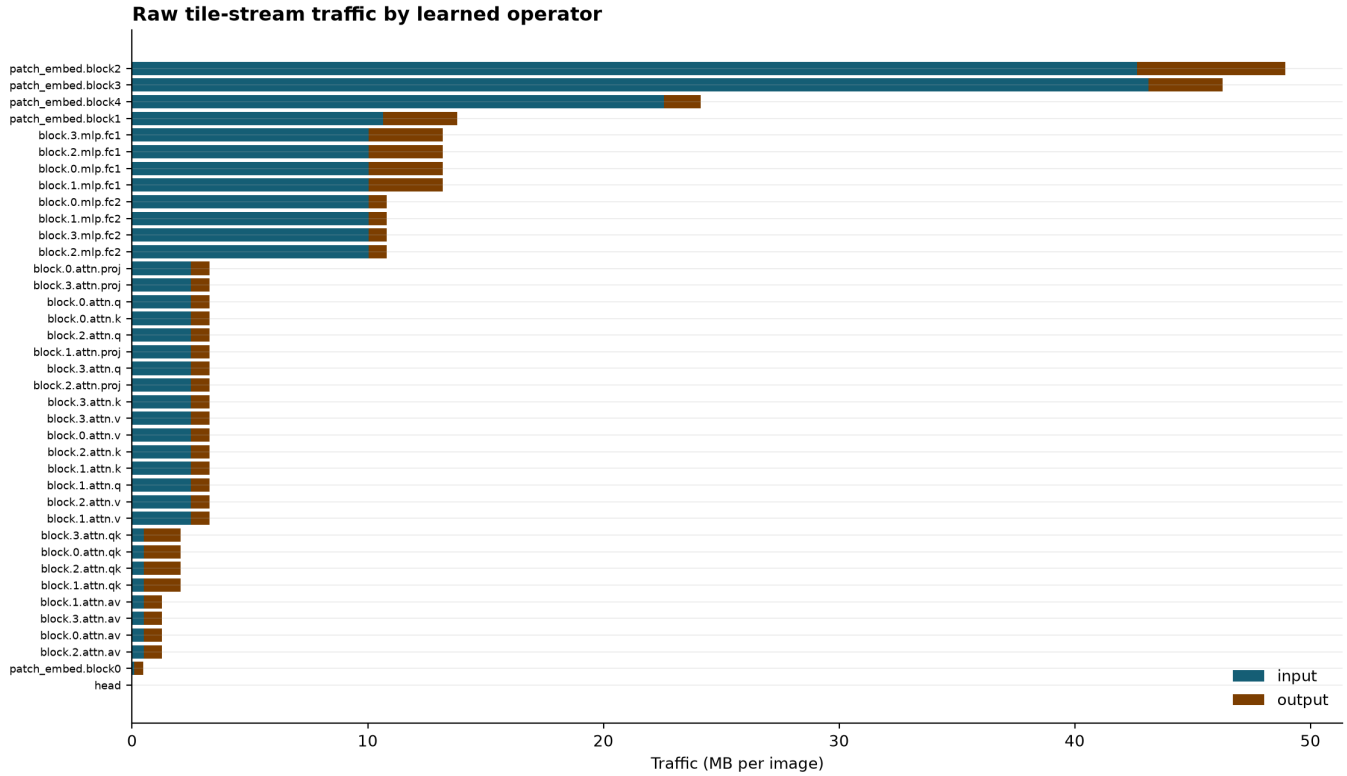
17 Tile commands by learned operator

Evidence: execution manifest generated from exported tensor shapes



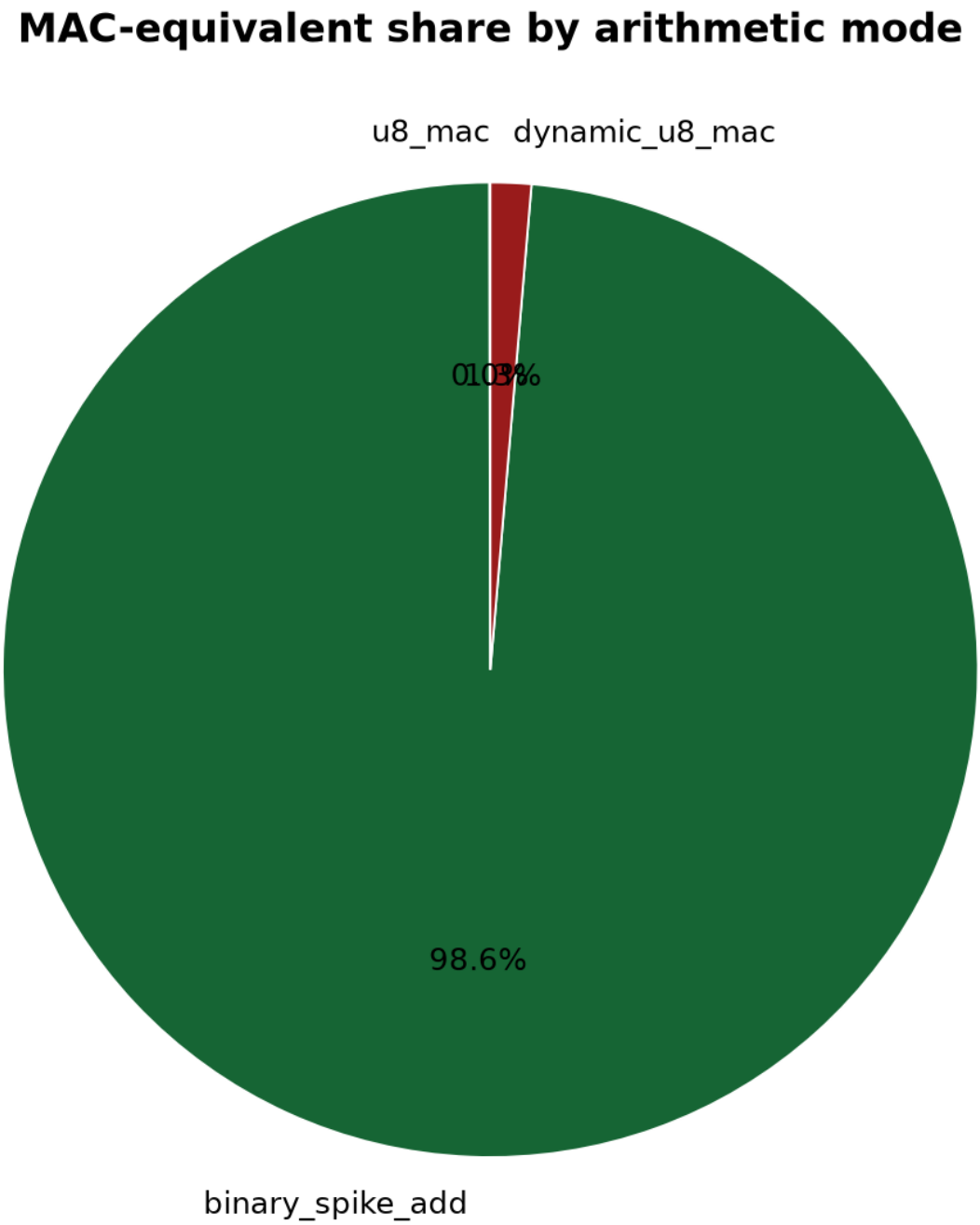
18 Operator stream traffic

Evidence: execution manifest; excludes inter-layer DDR spill



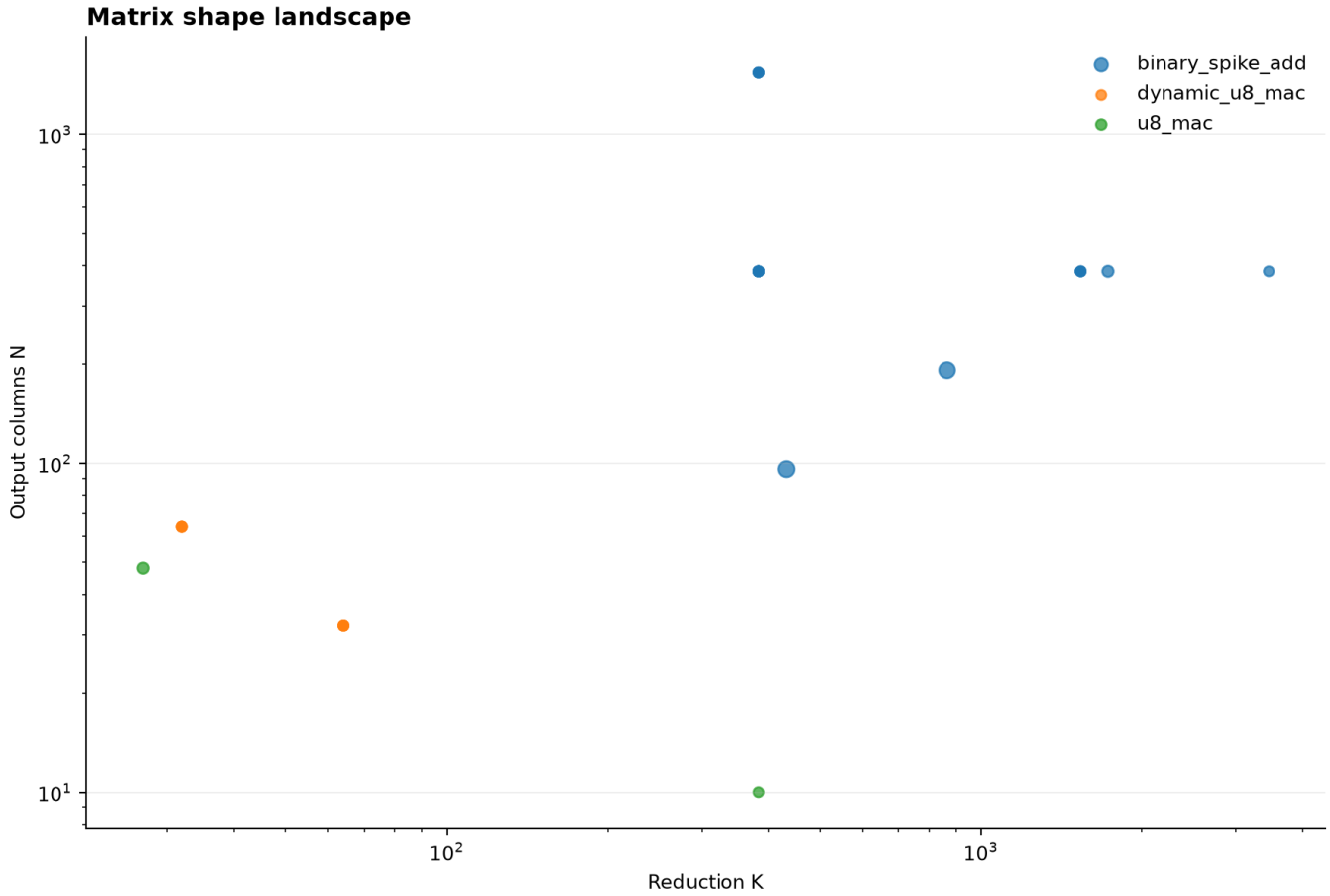
19 Arithmetic mode share

Evidence: execution manifest model-equivalent operation counts



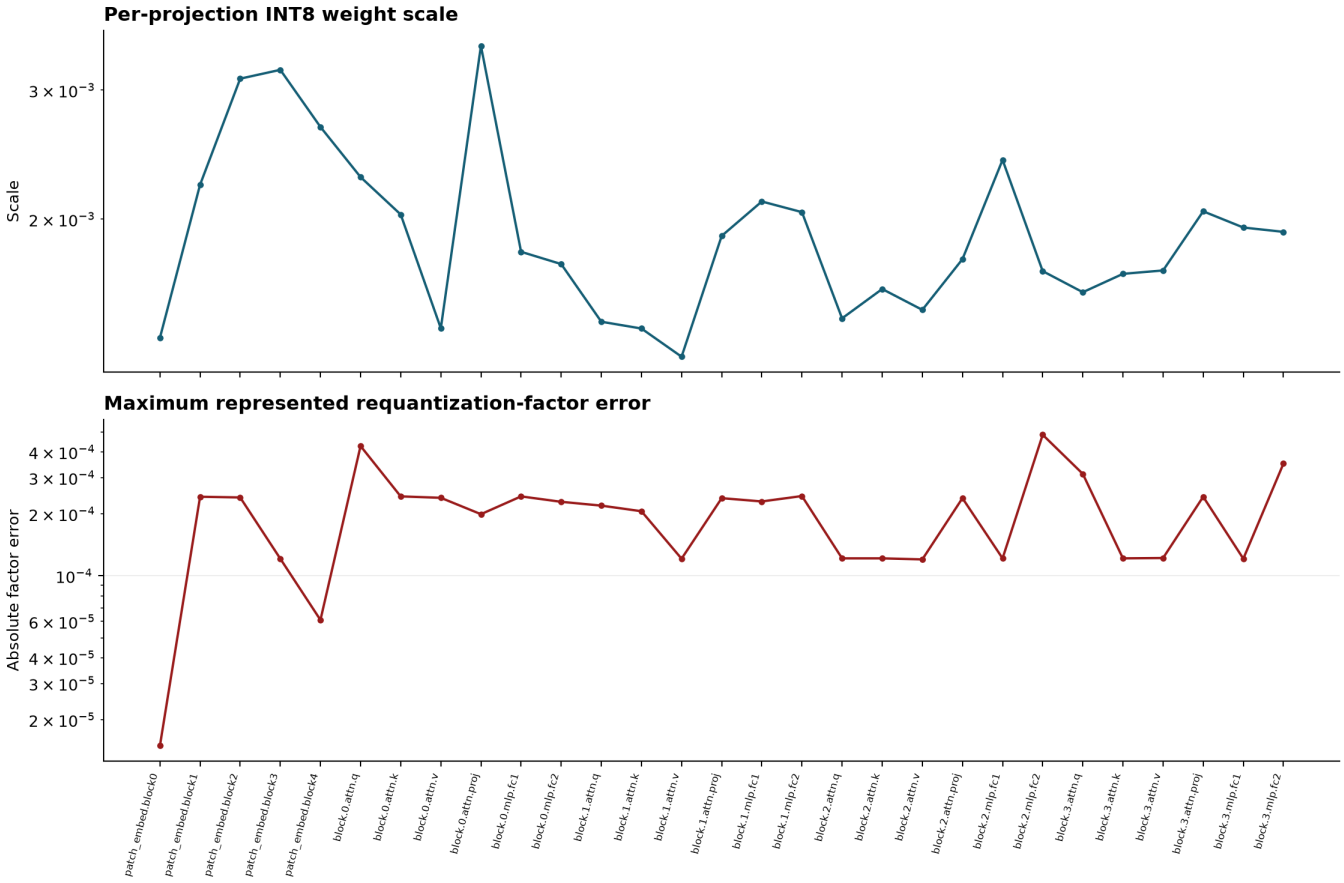
20 Matrix shape landscape

Evidence: execution manifest; marker area tracks output-row count



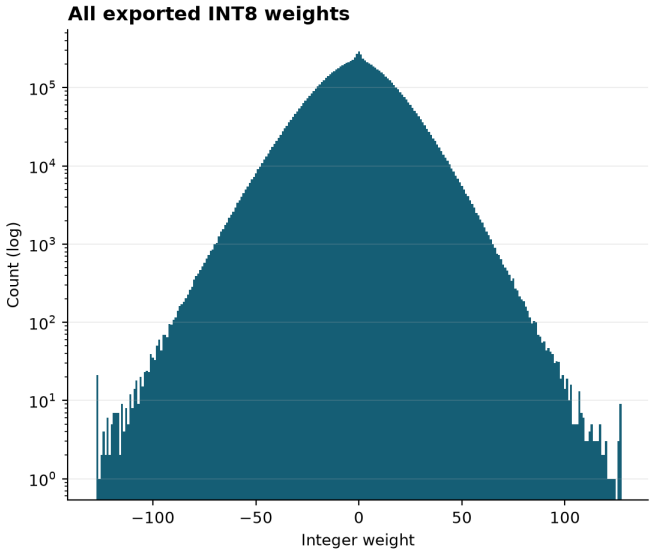
21 Projection quantization diagnostics

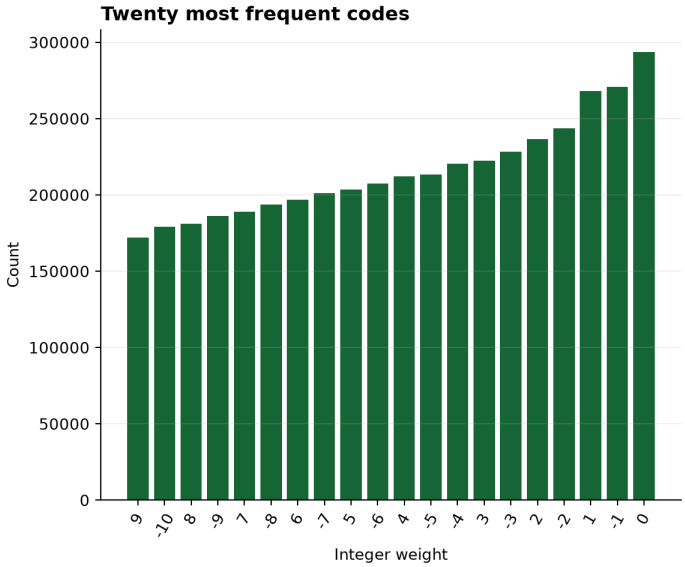
Evidence: hashed hardware parameter manifest



22 Exported weight distribution

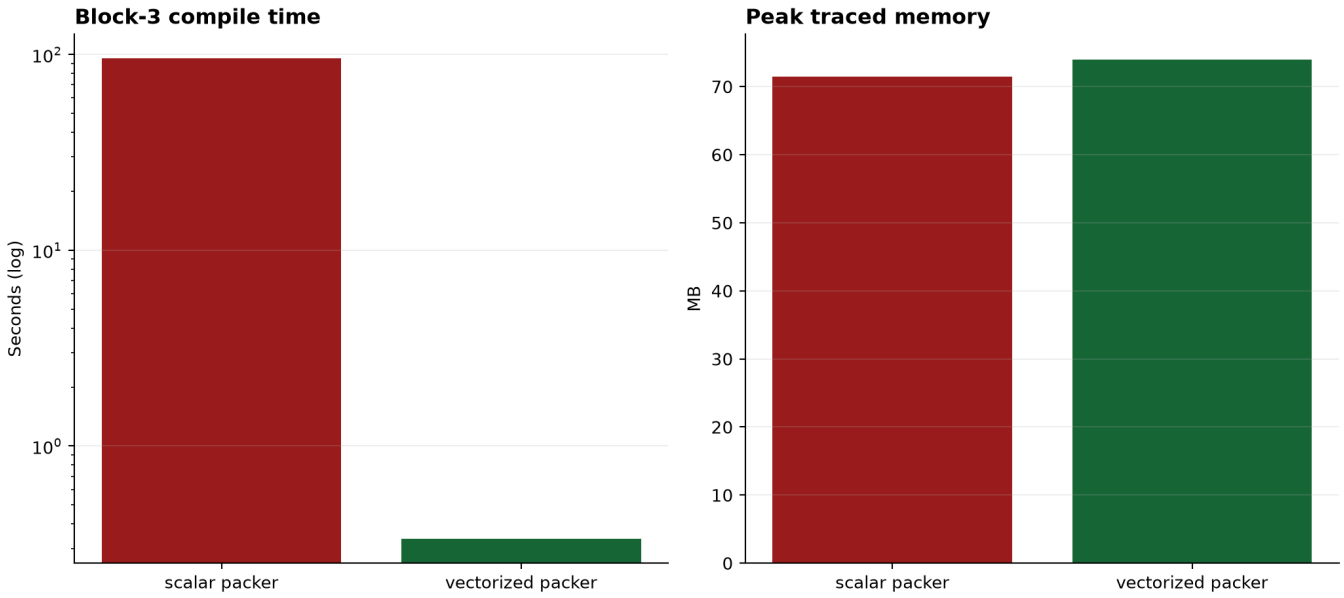
Evidence: direct section views from the SHA-256-checked parameter binary





23 Largest-layer compiler benchmark

Evidence: recorded scalar baseline and outputs/benchmarks/compiler_block3_memory.json



24 Full-backbone training schedule and numeric contract

Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\training\spikingformer_training_coverage.json

Full-Backbone Training Schedule

```
Target: complete exported integer inference graph

Execution operators covered: 43/43; uncovered 0

Learned affine groups / dynamic attention products: 30 / 8

Trainable deployed parameter elements: 9,295,978

Backward partial tiles / signed multiplications: 14,742,724 / 7,474,847,252

Fabric SGDW update commands: 1,161,998

Quantizer derivative: straight-through estimator with fixed exported scales

LIF derivative: SpikingJelly sigmoid surrogate alpha=4, hard reset, detach_reset=True

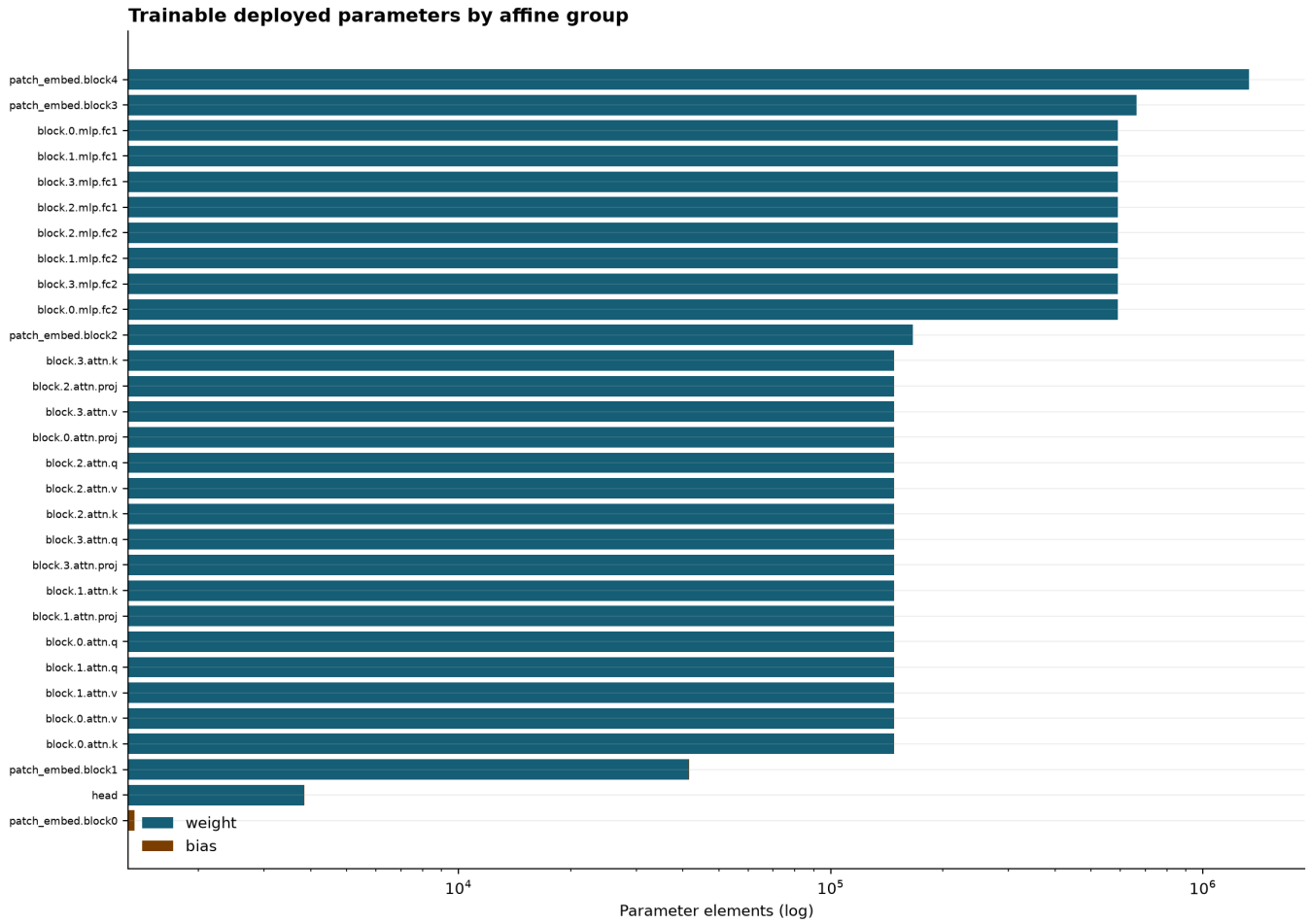
Temporal schedule: four-step BPTT in reverse timestep order through cached membrane traces

Optimizer: signed fixed-point SGDW with momentum; weight decay is applied to the update direction and excluded from retained velocity

Parameter residency: PYNQ DDR master tensors; arithmetic and updates in programmable logic
```

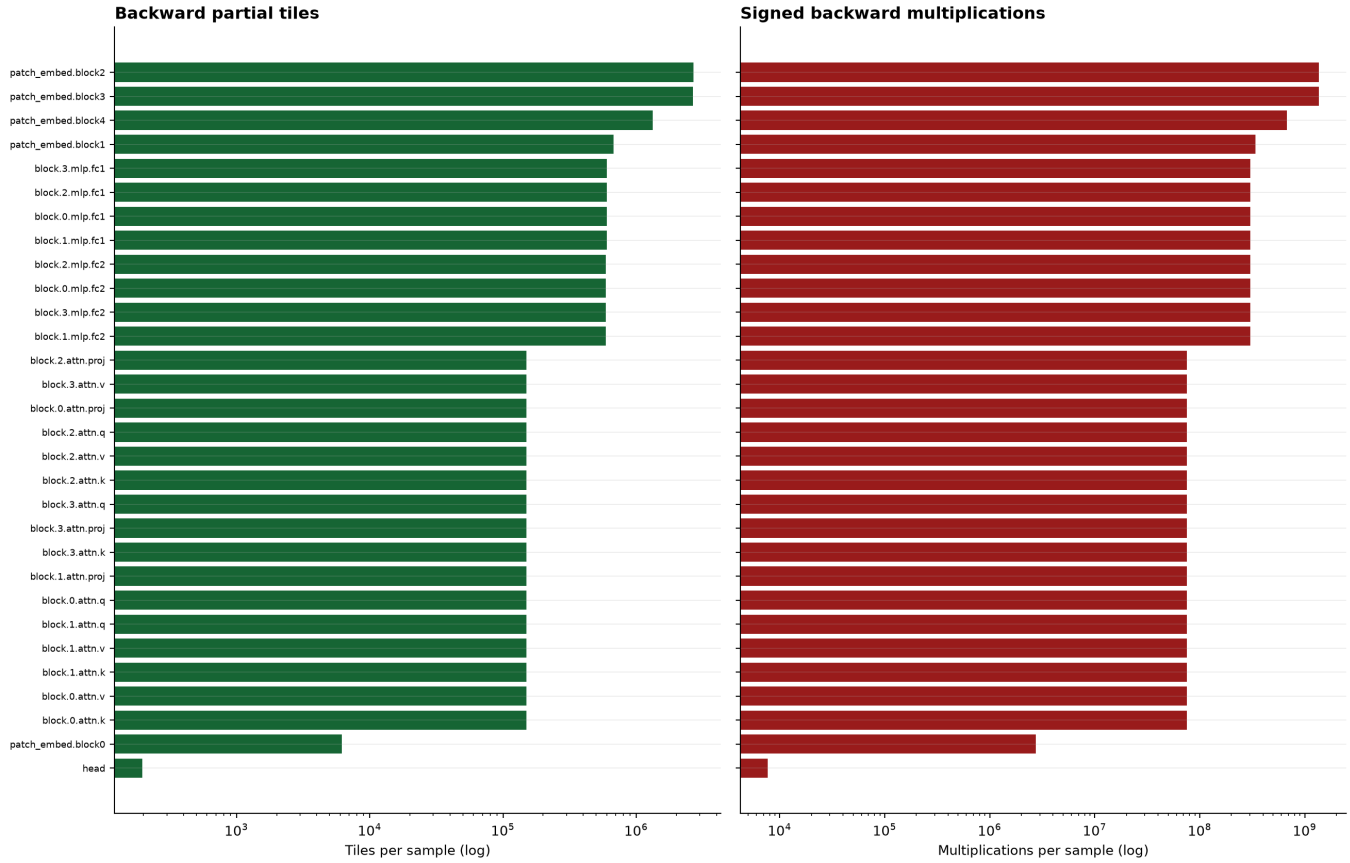
25 Trainable deployed parameter distribution

Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\training\spikingformer_training_coverage.json



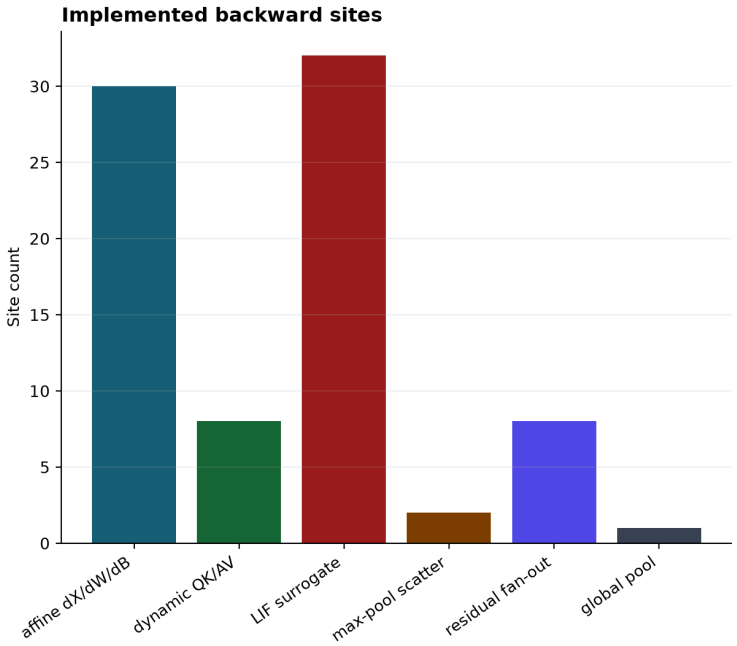
26 Exact learned-layer backward work

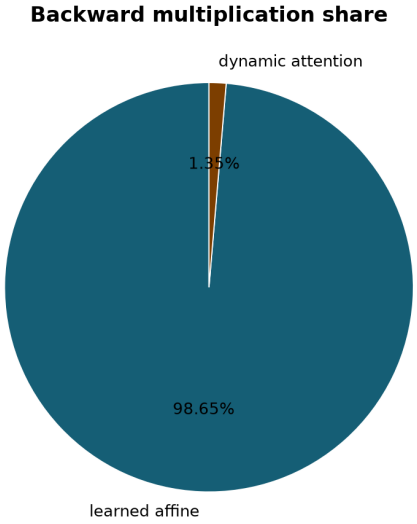
Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\training\spikingformer_training_coverage.json



27 Backward operator coverage and work composition

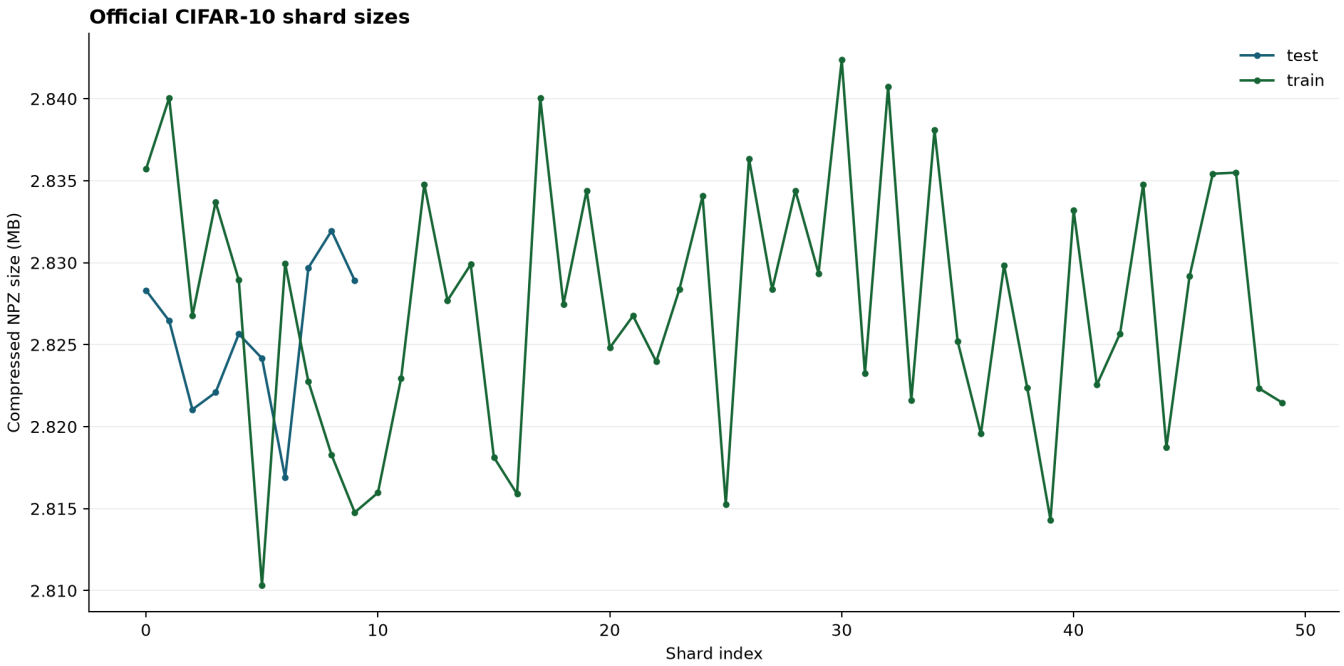
Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\training\spikingformer_training_coverage.json





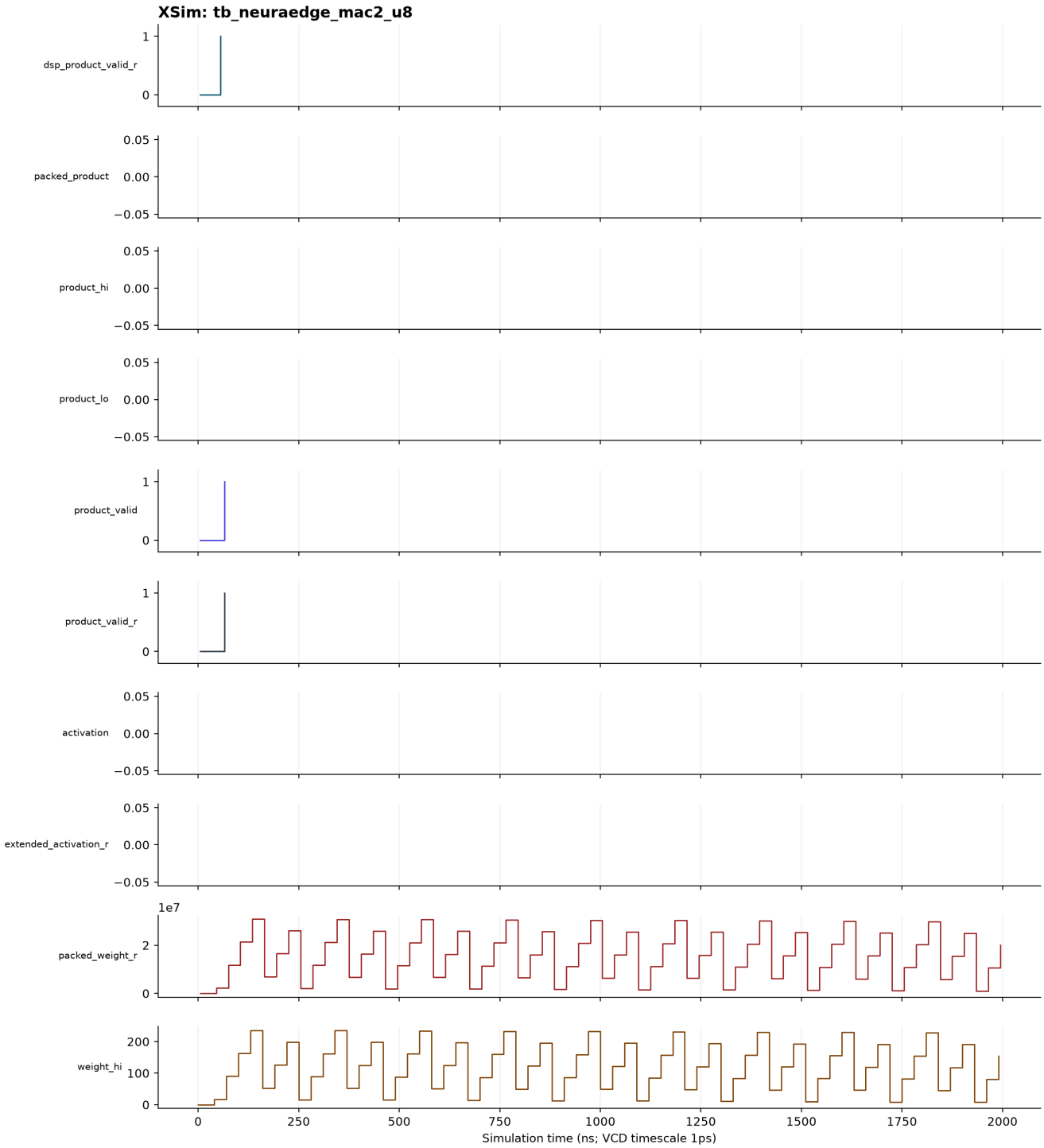
28 Deterministic CIFAR-10 board-training shards

Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\cifar10_shards\cifar10_shards_manifest.json



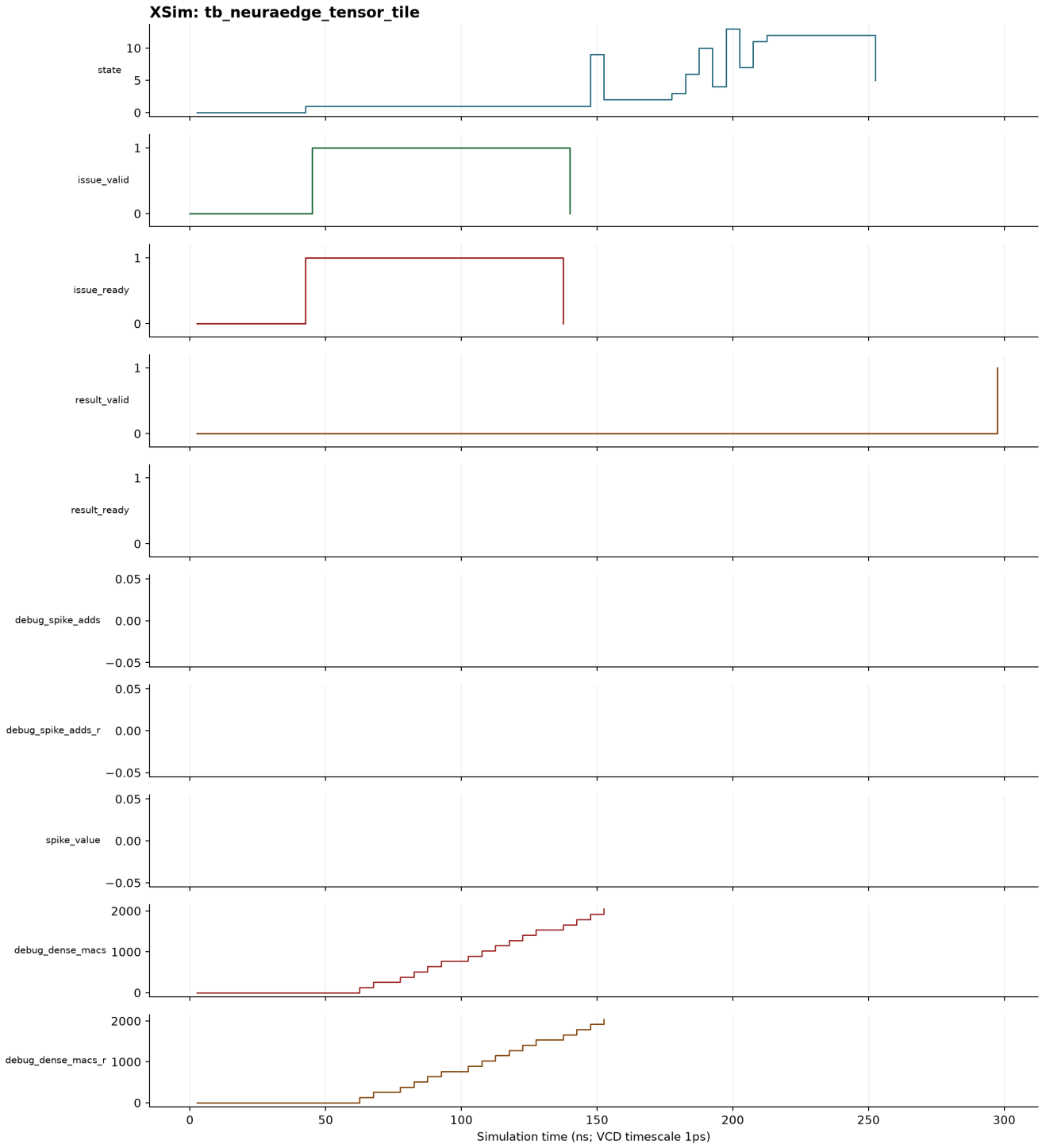
29 Waveform: tb_neuraedge_mac2_u8

Evidence: Vivado 2026.1 XSim VCD, requested window 2 us; signals: dsp_product_valid_r, packed_product, product_hi, product_lo, product_valid, product_valid_r, activation, extended_activation_r, packed_weight_r, weight_hi



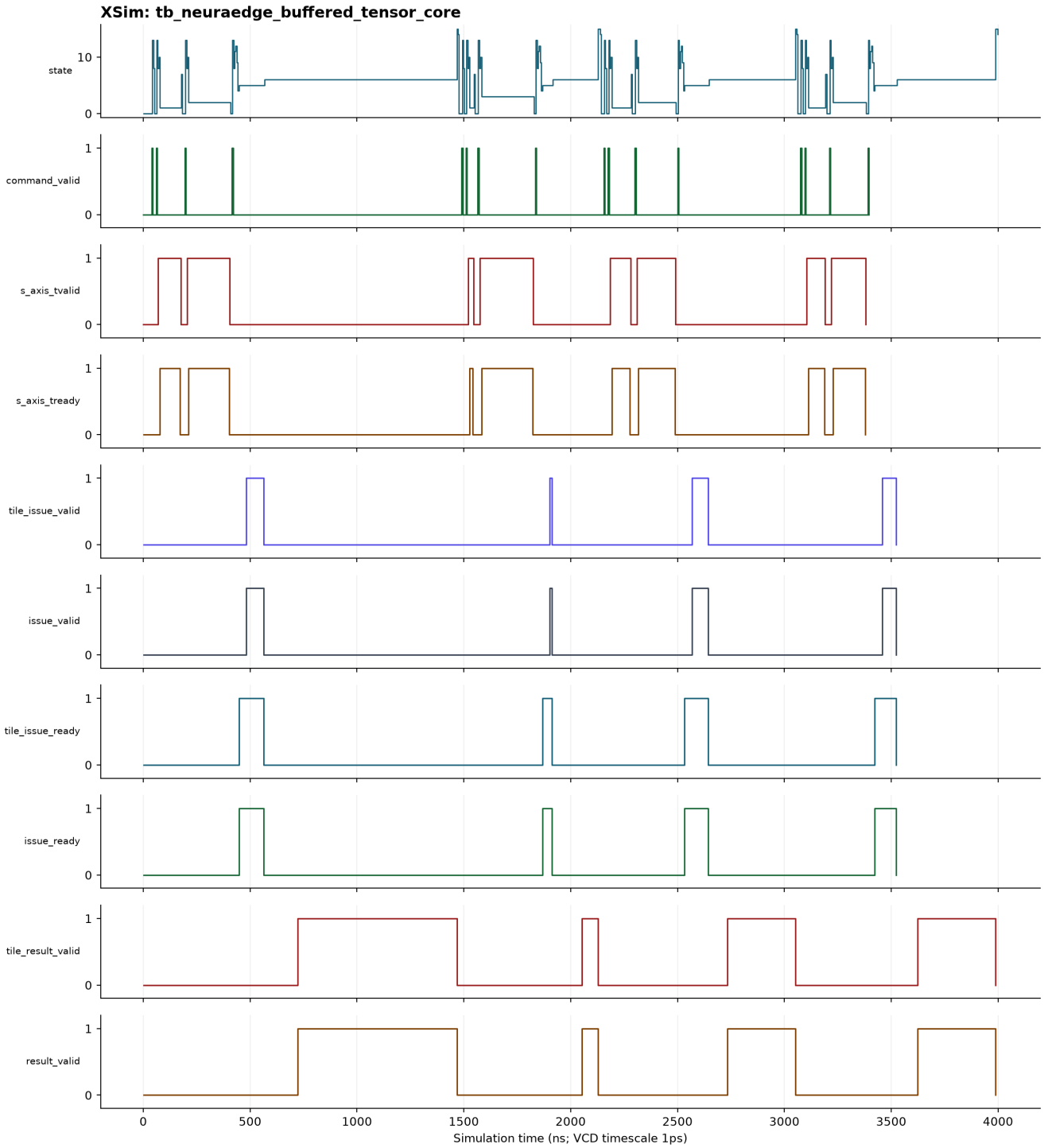
30 Waveform: tb_neuraedge_tensor_tile

Evidence: Vivado 2026.1 XSim VCD, requested window 3 us; signals: state, issue_valid, issue_ready, result_valid, result_ready, debug_spike_adds, debug_spike_adds_r, spike_value, debug_dense_macs, debug_dense_macs_r



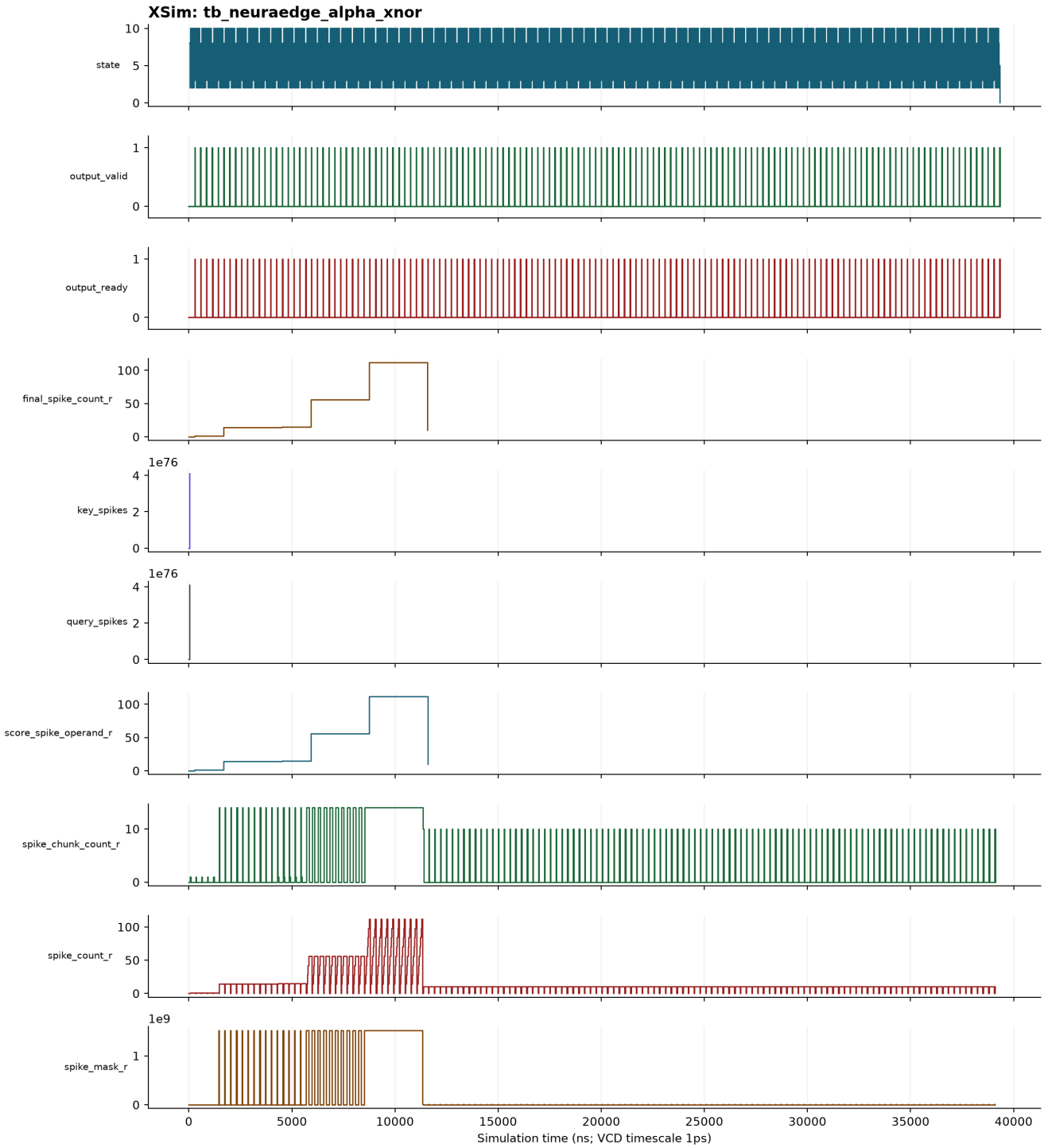
31 Waveform: tb_neuraedge_buffered_tensor_core

Evidence: Vivado 2026.1 XSim VCD, requested window 4 us; signals: state, command_valid, s_axis_tvalid, s_axis_tready, tile_issue_valid, issue_valid, tile_issue_ready, issue_ready, tile_result_valid, result_valid



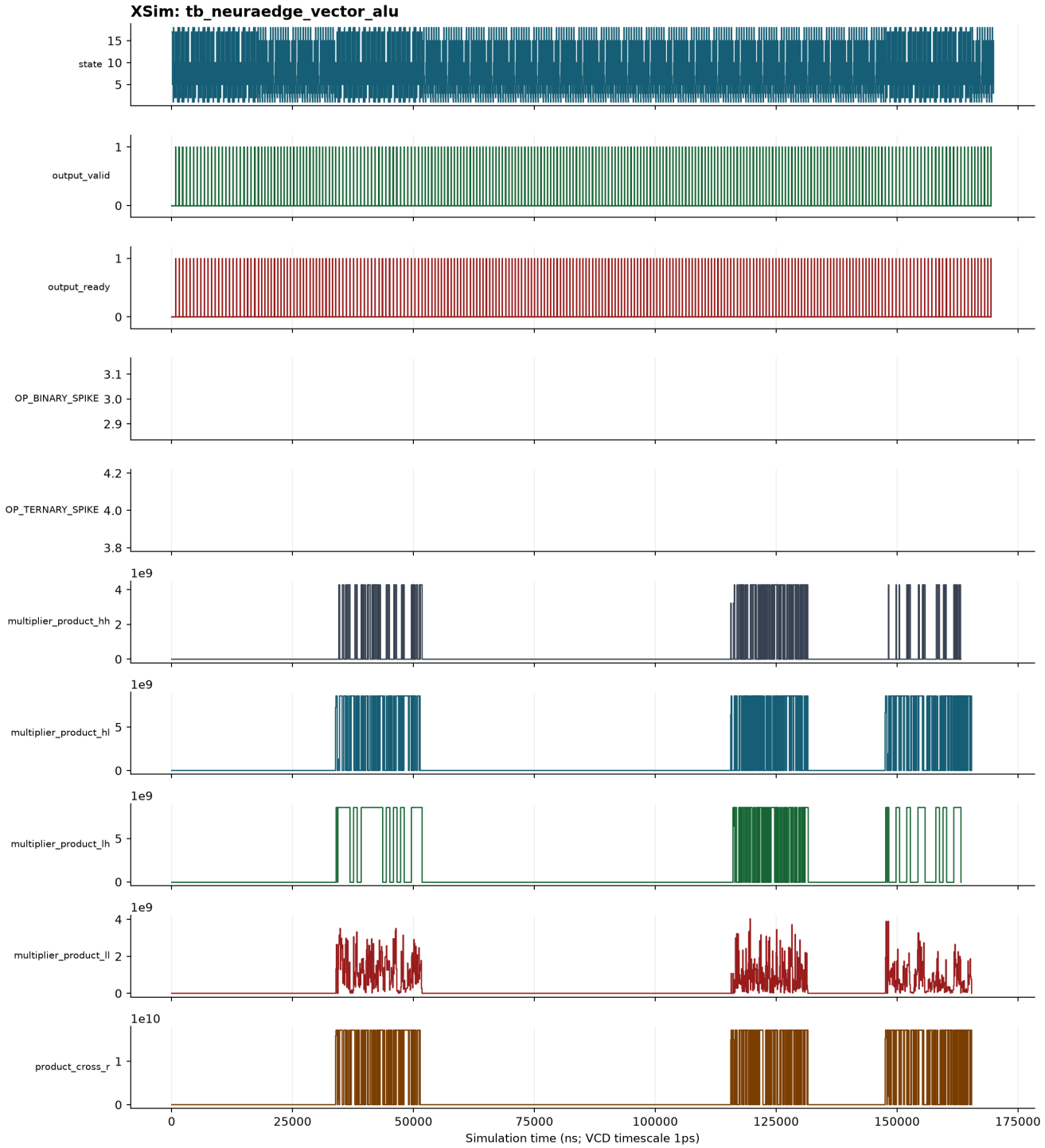
32 Waveform: tb_neuraedge_alpha_xnor

Evidence: Vivado 2026.1 XSim VCD, requested window 40 us; signals: state, output_valid, output_ready, final_spike_count_r, key_spikes, query_spikes, score_spike_operand_r, spike_chunk_count_r, spike_count_r, spike_mask_r



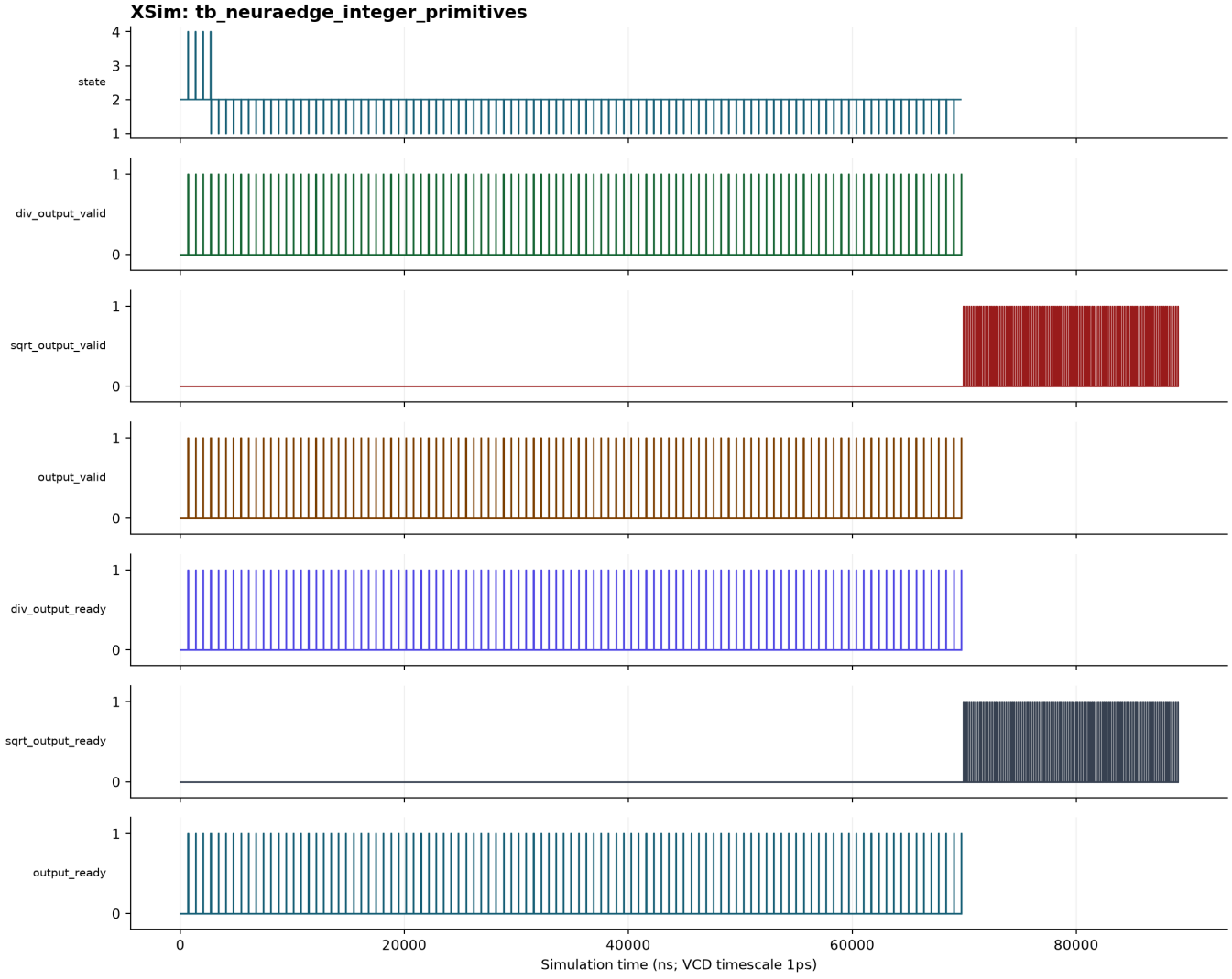
33 Waveform: tb_neuraedge_vector_alu

Evidence: Vivado 2026.1 XSim VCD, requested window 170 us; signals: state, output_valid, output_ready, OP_BINARY_SPIKE, OP_TERNARY_SPIKE, multiplier_product_hh, multiplier_product_hl, multiplier_product_lh, multiplier_product_ll, product_cross_r



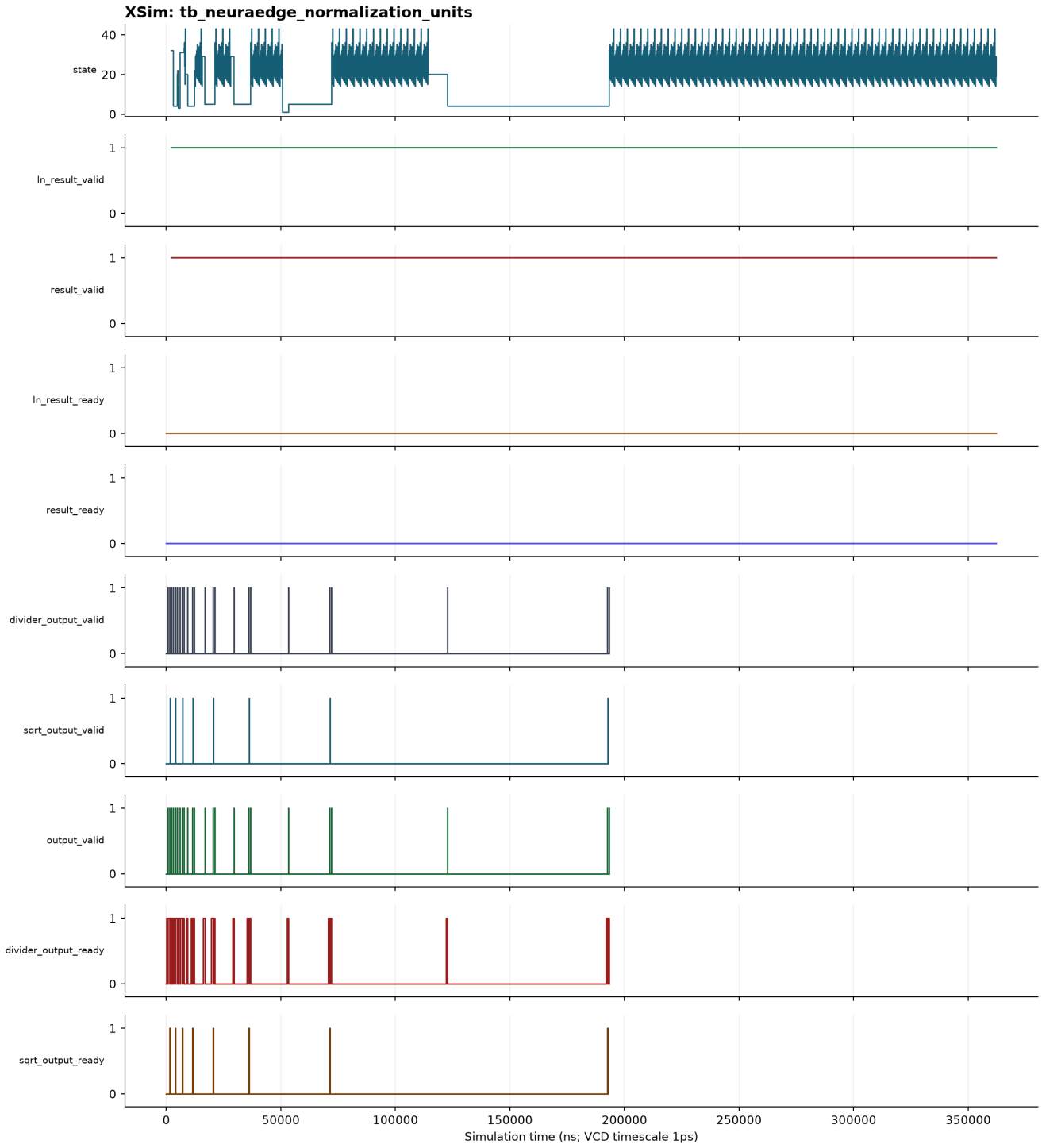
34 Waveform: tb_neuraedge_integer_primitives

Evidence: Vivado 2026.1 XSim VCD, requested window 90 us; signals: state, div_output_valid, sqrt_output_valid, output_valid, div_output_ready, sqrt_output_ready, output_ready



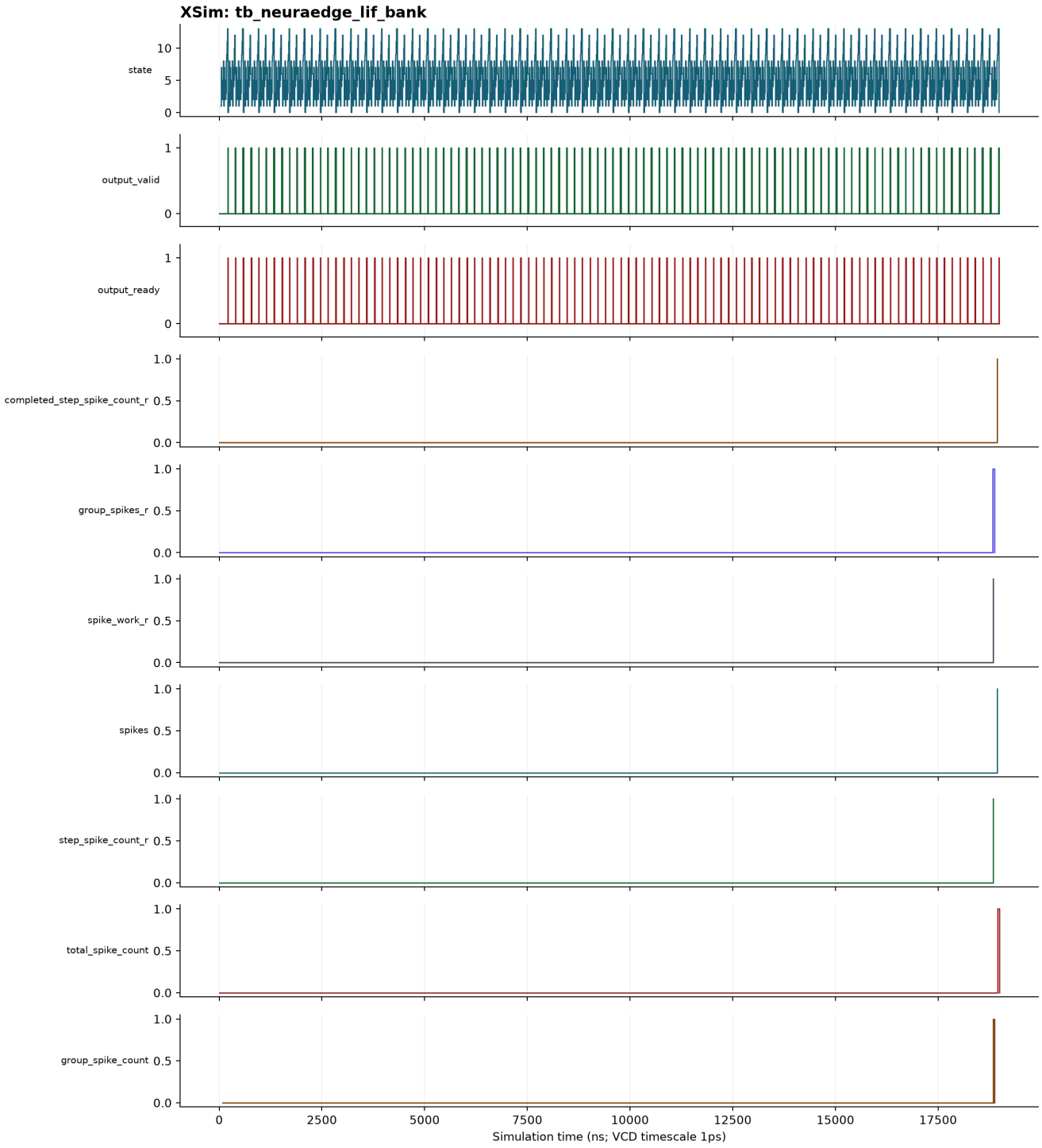
35 Waveform: tb_neuraedge_normalization_units

Evidence: Vivado 2026.1 XSim VCD, requested window 405 us; signals: state, ln_result_valid, result_valid, ln_result_ready, result_ready, divider_output_valid, sqrt_output_valid, output_valid, divider_output_ready, sqrt_output_ready



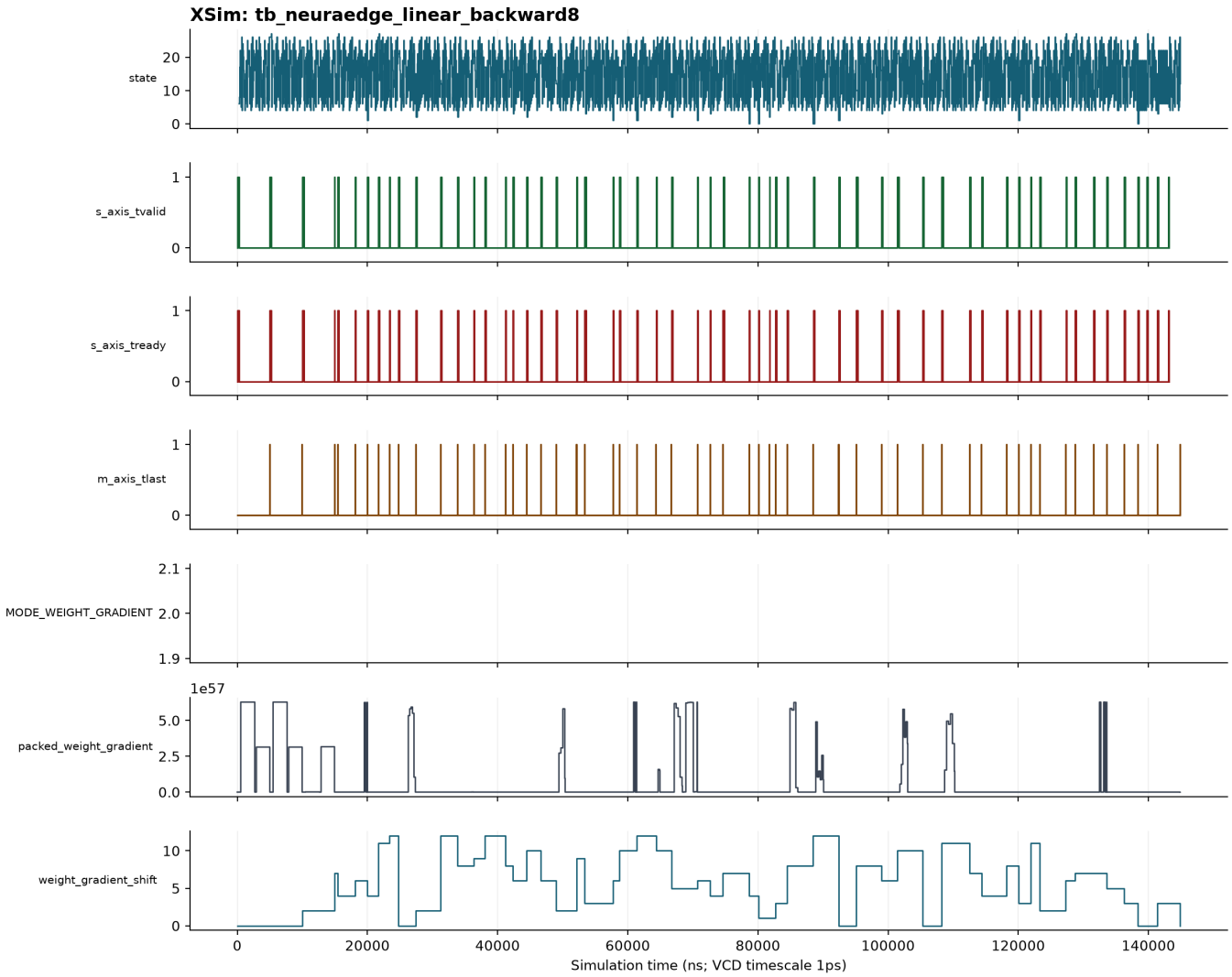
36 Waveform: tb_neuraedge_lif_bank

Evidence: Vivado 2026.1 XSim VCD, requested window 20 us; signals: state, output_valid, output_ready, completed_step_spike_count_r, group_spikes_r, spike_work_r, spikes, step_spike_count_r, total_spike_count, group_spike_count



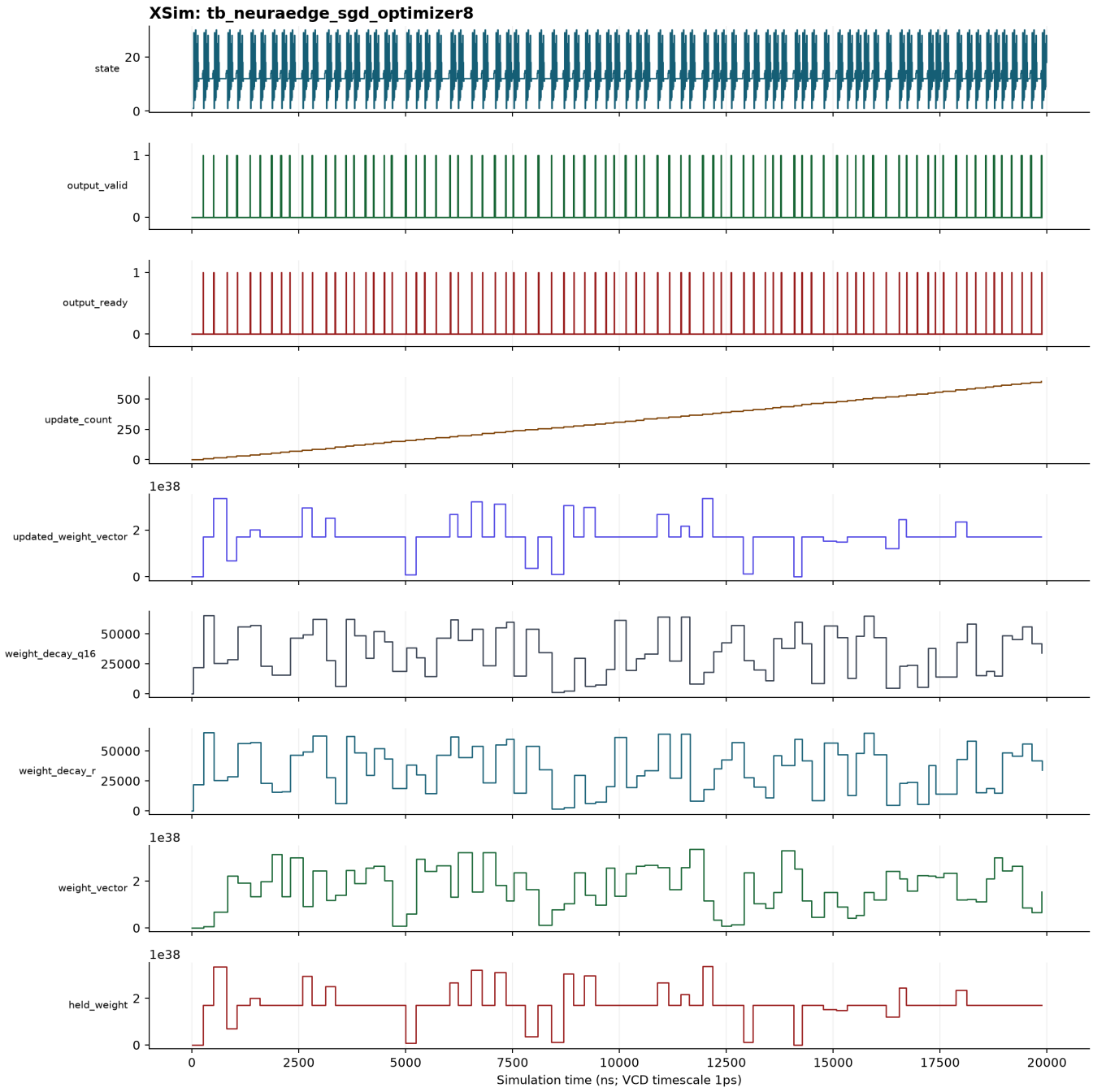
37 Waveform: tb_neuraedge_linear_backward8

Evidence: Vivado 2026.1 XSim VCD, requested window 1 ms; signals: state, s_axis_tvalid, s_axis_tready, m_axis_tlast, MODE_WEIGHT_GRADIENT, packed_weight_gradient, weight_gradient_shift



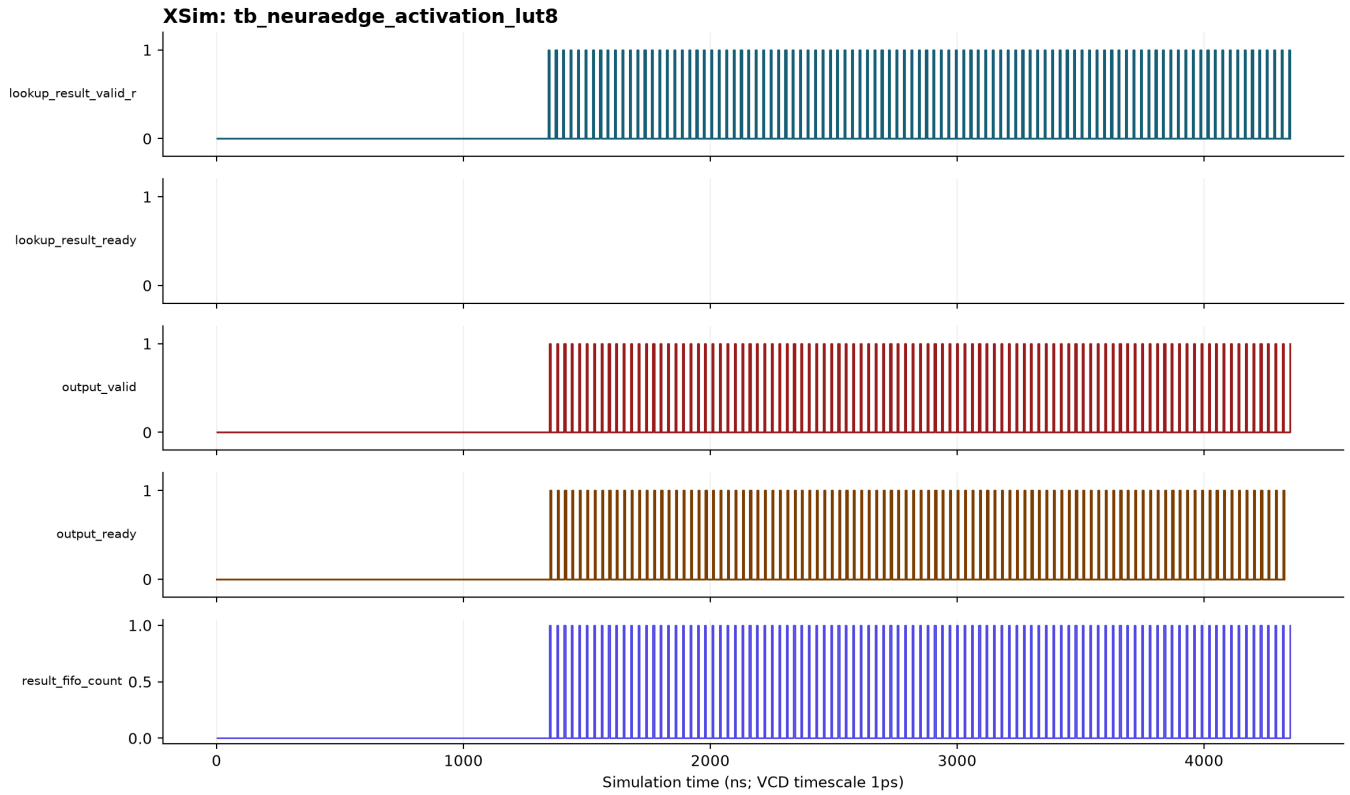
38 Waveform: tb_neuraedge_sgd_optimizer8

Evidence: Vivado 2026.1 XSim VCD, requested window 20 us; signals: state, output_valid, output_ready, update_count, updated_weight_vector, weight_decay_q16, weight_decay_r, weight_vector, held_weight



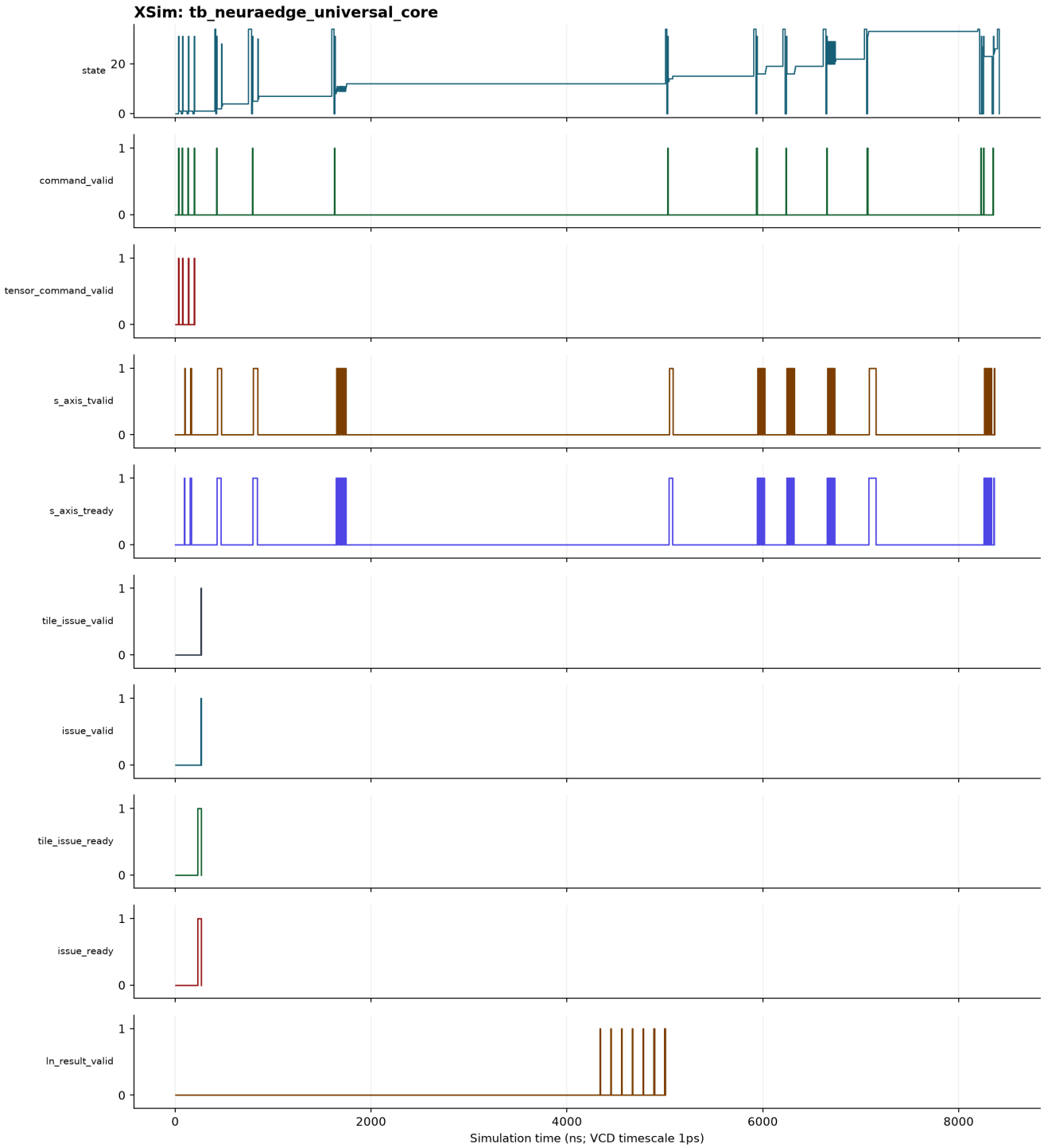
39 Waveform: tb_neuraedge_activation_lut8

Evidence: Vivado 2026.1 XSim VCD, requested window 20 us; signals: lookup_result_valid_r, lookup_result_ready, output_valid, output_ready, result_fifo_count



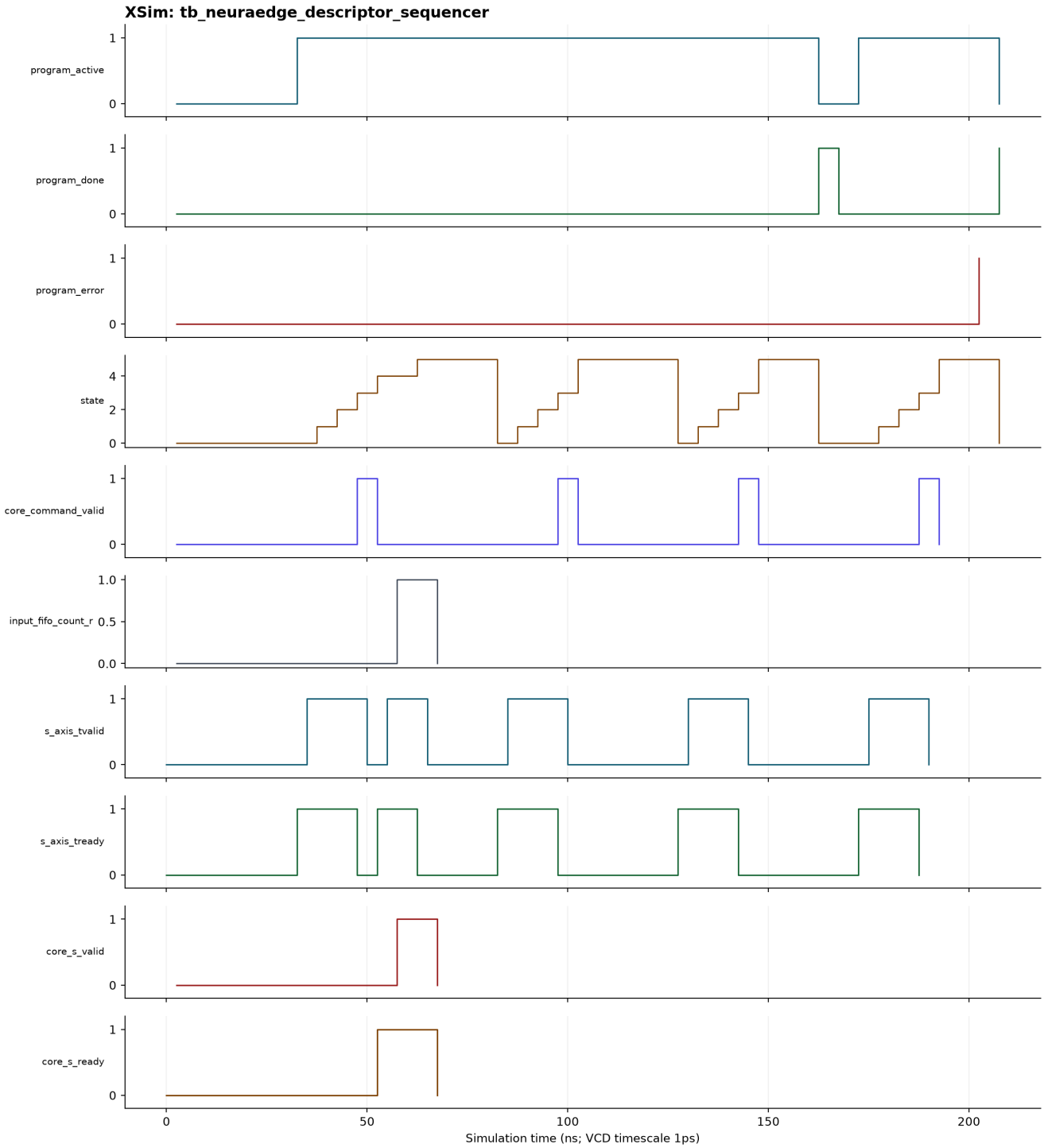
40 Waveform: tb_neuraedge_universal_core

Evidence: Vivado 2026.1 XSim VCD, requested window 50 us; signals: state, command_valid, tensor_command_valid, s_axis_tvalid, s_axis_tready, tile_issue_valid, issue_valid, tile_issue_ready, issue_ready, ln_result_valid



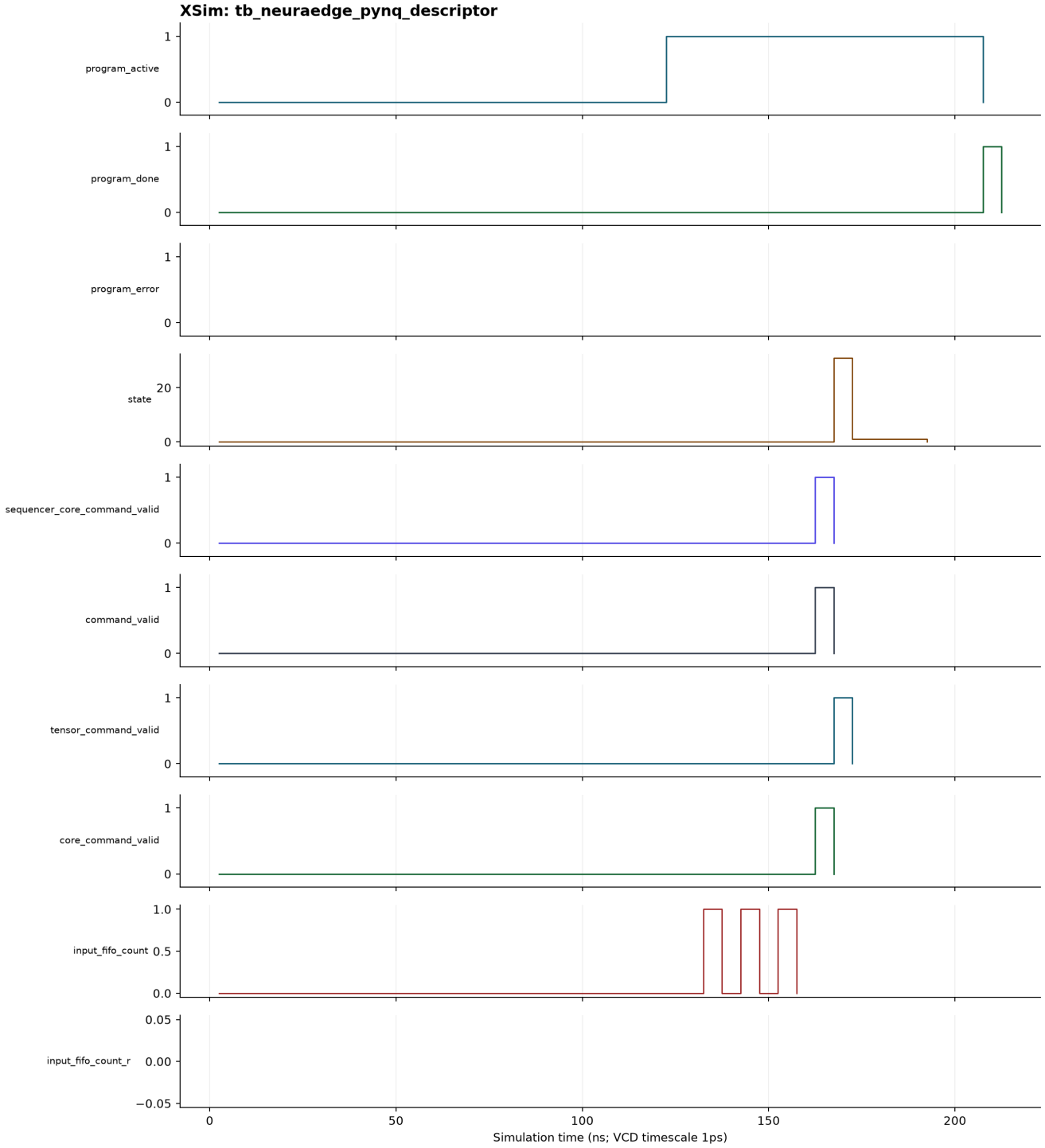
41 Waveform: tb_neuraedge_descriptor_sequencer

Evidence: Vivado 2026.1 XSim VCD, requested window 2 us; signals: program_active, program_done, program_error, state, core_command_valid, input_fifo_count_r, s_axis_tvalid, s_axis_tready, core_s_valid, core_s_ready



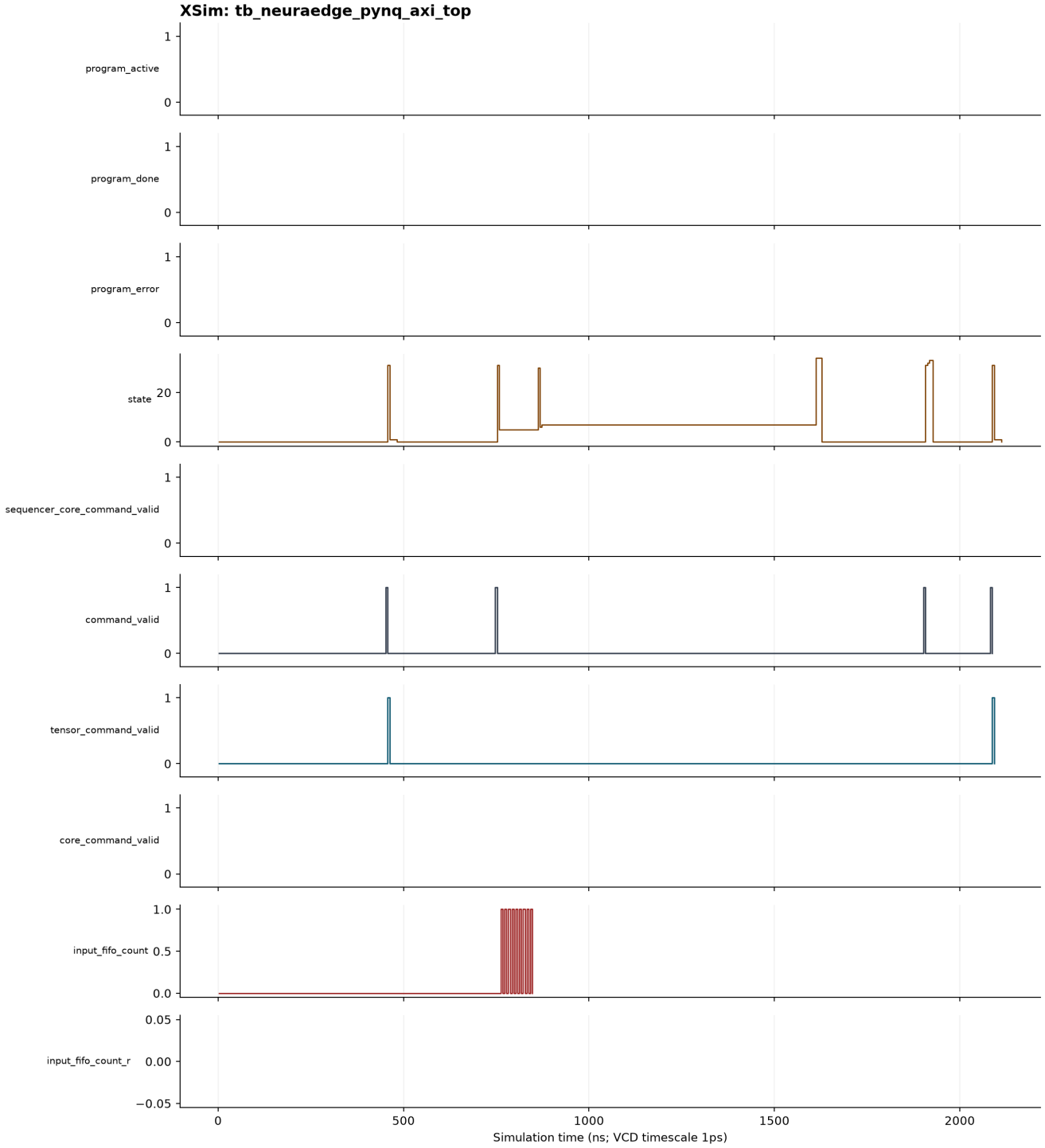
42 Waveform: tb_neuraedge_pynq_descriptor

Evidence: Vivado 2026.1 XSim VCD, requested window 2 us; signals: program_active, program_done, program_error, state, sequencer_core_command_valid, command_valid, tensor_command_valid, core_command_valid, input_fifo_count, input_fifo_count_r



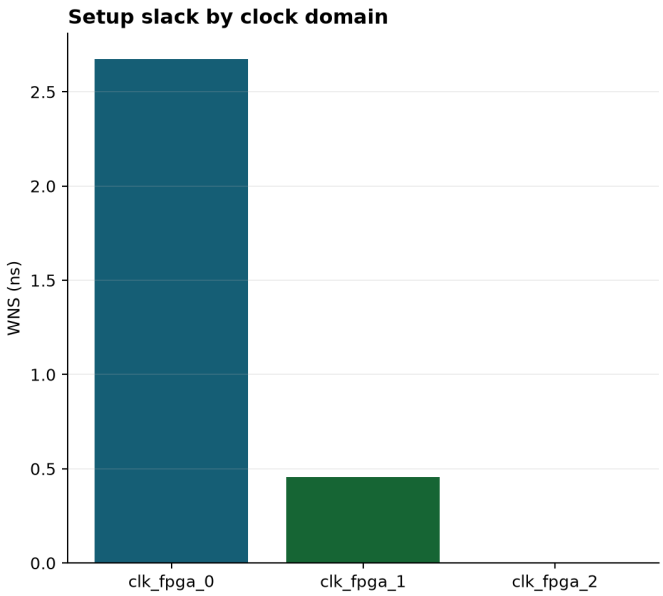
43 Waveform: tb_neuraedge_pynq_axi_top

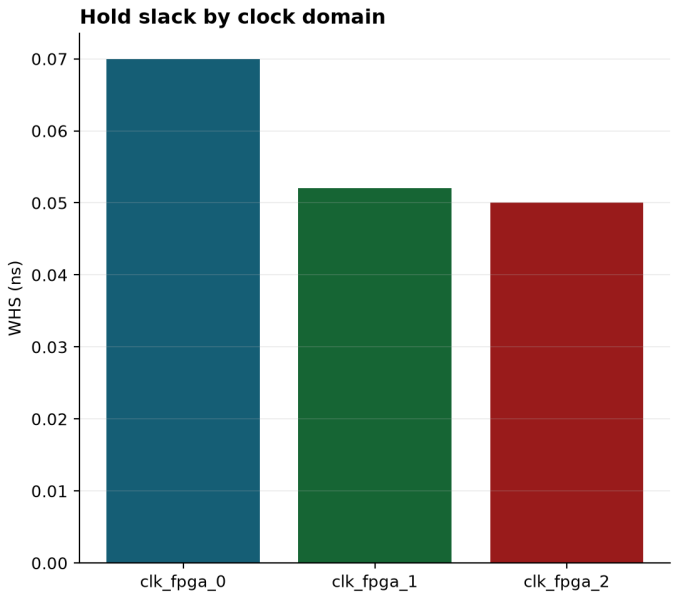
Evidence: Vivado 2026.1 XSim VCD, requested window 5 us; signals: program_active, program_done, program_error, state, sequencer_core_command_valid, command_valid, tensor_command_valid, core_command_valid, input_fifo_count, input_fifo_count_r



44 Routed timing by clock domain

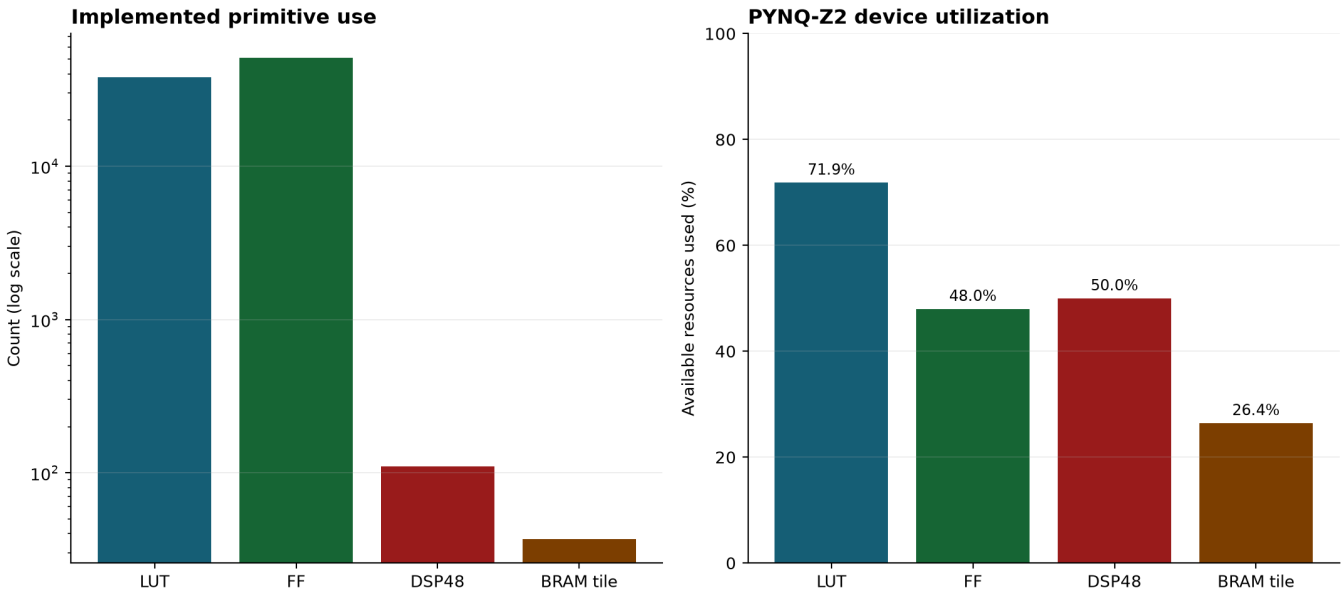
Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\pynq_full\full_routed_timing_summary.rpt





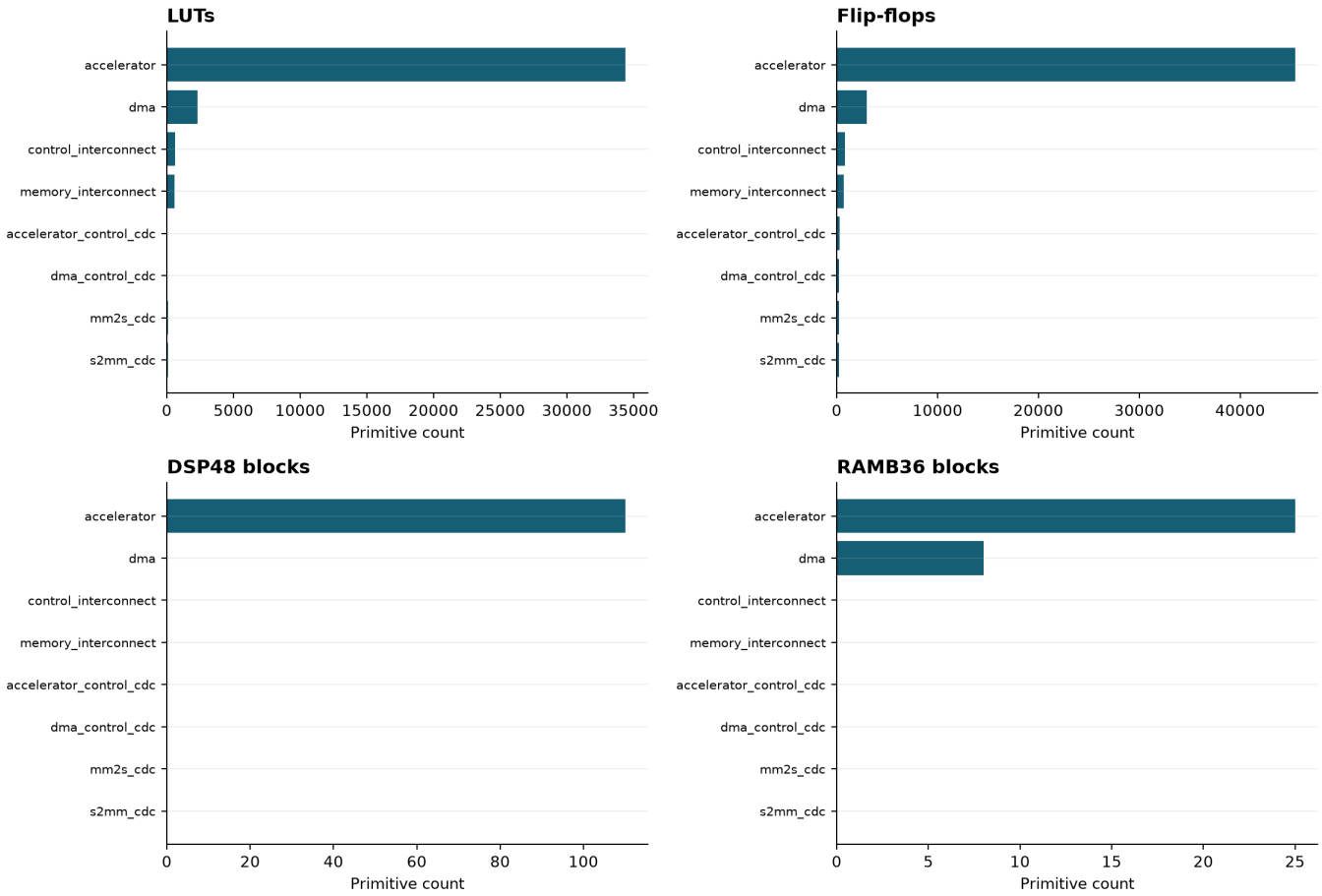
45 Full PYNQ routed resource utilization

Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\pynq_full\full_routed_utilization_hierarchical.rpt



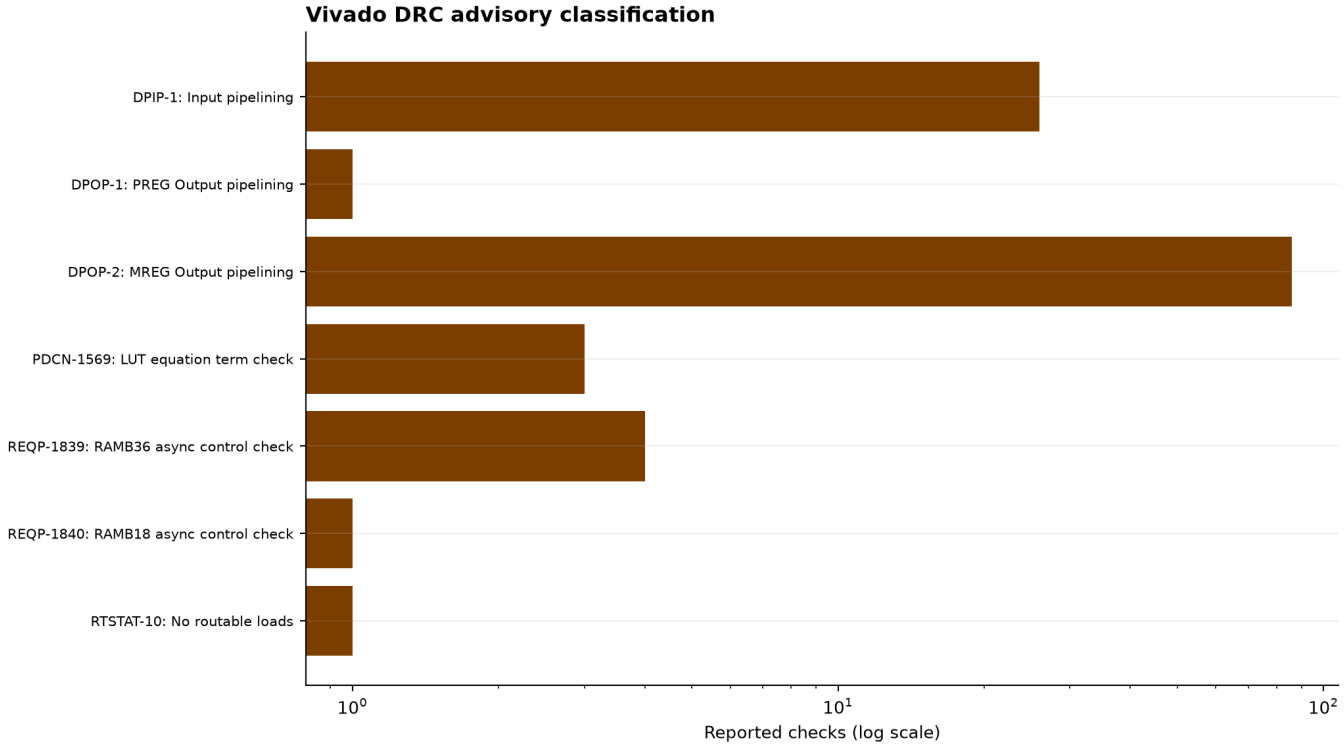
46 Resource use by full-system hierarchy

Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\pynq_full\full_routed_utilization_hierarchical.rpt



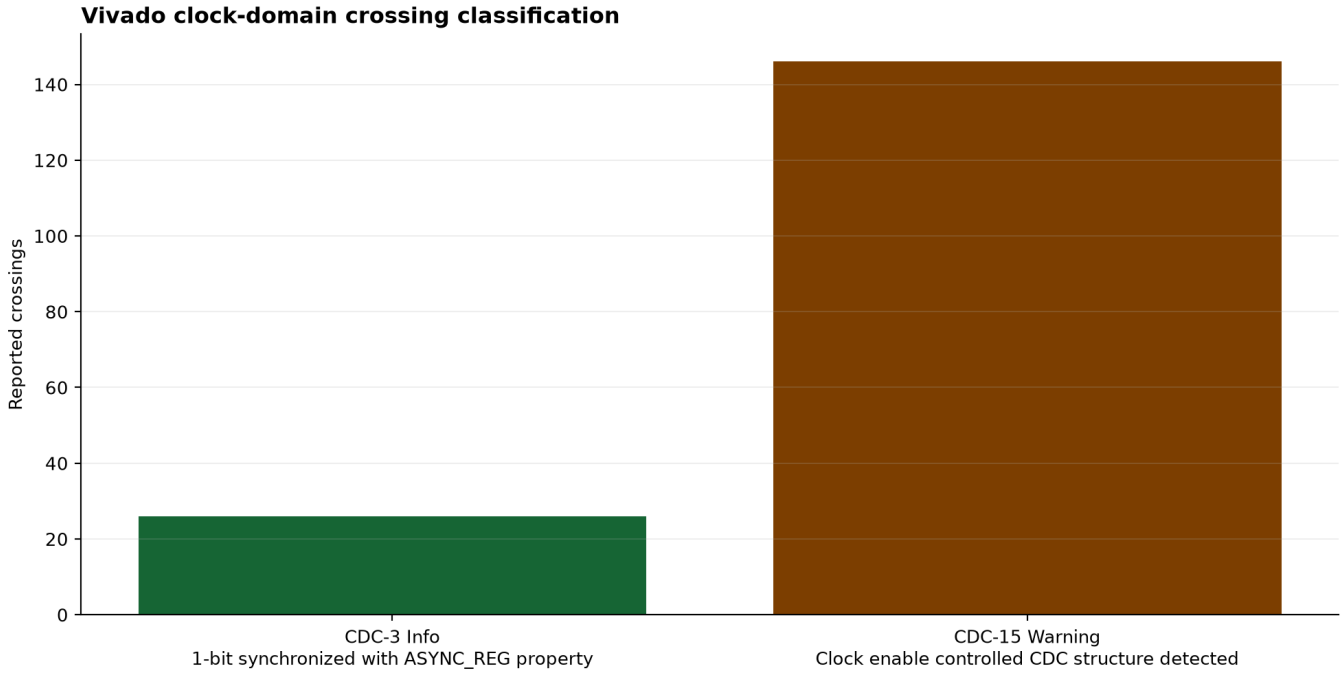
47 DRC warning classification

Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\pynq_full\full_drc.rpt



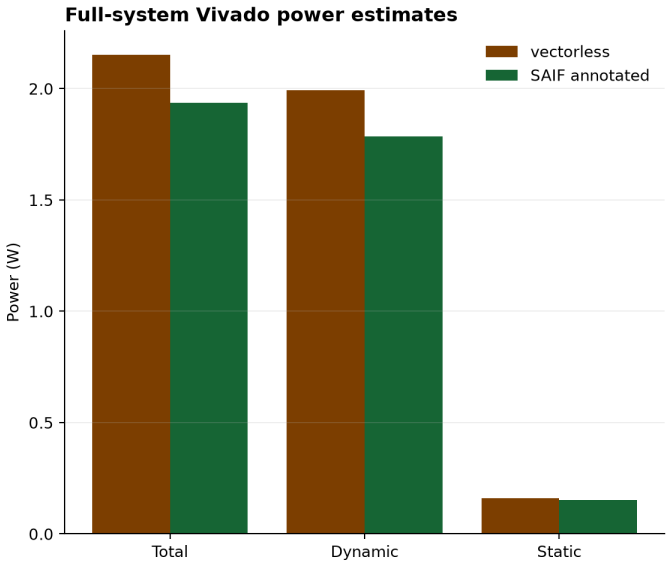
48 Clock-domain crossing classification

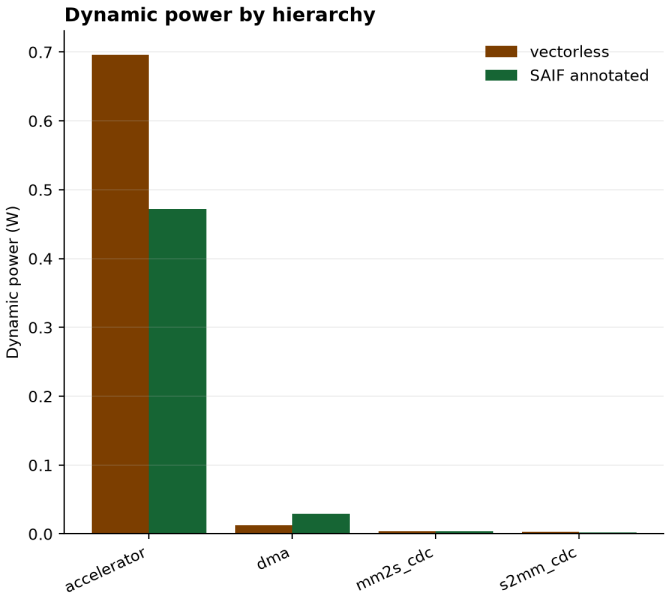
Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\pynq_full\full_cdc.rpt



49 Vectorless and SAIF-annotated power estimates

Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\pynq_full\full_vectorless_power.rpt; C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\power_activity_full\vivado_reports\routed_saif_power.rpt; SAIF match 7% (9164/123250)





50 Power confidence and annotation coverage

Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\power_activity_full\vivado_reports\routed_saif_power.rpt

Power Evidence and Confidence Boundary

```
Vectorless: total 2.152 W, dynamic 1.993 W, confidence Medium

SAIF: total 1.937 W, dynamic 1.785 W, confidence Medium

SAIF implementation nets matched: 7%   (9164/123250)

Captured tests do not initialize every memory location; Vivado reports X-state warnings for untouched storage.

These are tool estimates for the routed design, not measurements from a PYNQ-Z2 power rail.
```

51 Routed timing path distribution

Evidence: C:\Users\Ross\Downloads\neuraedge–main\neuraedge–main\neuraedge_universal\outputs\pynq_full\full_routed_timing_paths_250.rpt

